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ATFA: Topology-aware aggregation with Laplacian-based heat diffusion for federated learning

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ABSTRACT

Federated Learning (FL) faces significant performance degradation in non-IID scenarios due to divergent optimization directions among local models. Conventional aggregation techniques, which are mostly based on Euclidean averaging, are susceptible to stochastic optimization and frequently fail to capture the intrinsic manifold structure of client updates. To address these limitations, we propose the Adaptive Topology-aware Federated Aggregation (ATFA), which reformulates global aggregation as a topology-aware knowledge fusion process over a dynamically evolving client graph topology. Specifically, ATFA introduces historical trend information to capture the cross-round evolution of inter-client relationships. Based on this, a density-calibrated soft adjacency employs robust dispersion statistics to construct a sparse and adaptive client graph topology. ATFA then performs structured knowledge fusion through a heat diffusion derived from the normalized Laplacian, enabling model information to fuse preferentially among strongly related clients. Extensive experiments on both standard benchmarks and three medical imaging benchmark datasets demonstrate that ATFA achieves competitive overall performance and notable improvements under various heterogeneity settings. Further experimental analyses confirm the contributions of the proposed mechanisms can improve aggregation stability and robustness.

1. Introduction

With the rapid development of the Internet of Things (IoT), billions of smart devices continuously generate massive amounts of perception data, providing unprecedented opportunities for training artificial intelligence models (Wazid et al., 2023). Centralizing the data distributed across edge terminals within a traditional machine learning framework poses significant privacy risks. Consequently, federated learning (FL) has emerged as a promising decentralized learning paradigm that enables collaborative model training among multiple institutions without the need to exchange raw data (Panda et al., 2025). In FL, clients perform local training and exchange model updates rather than raw data, thereby preserving data privacy while benefiting from collaborative learning. Federated averaging (FedAvg) is one of the most widely adopted FL algorithms (McMahan et al., 2016). In FedAvg, each participating client independently trains a local model, and the server aggregates the parameters in proportion to the size of the respective datasets to construct a global model. However, real-world client data are typically non-independent and identically distributed (non-IID). FedAvg is susceptible to client drift and update conflicts under statistical het-

erogeneity (Ahn et al., 2025; Huang et al., 2025). To mitigate the adverse effects of data heterogeneity, numerous FL methods have been developed (Jing et al., 2023), including adaptive aggregation strategies (Wang et al., 2025; Zhou et al., 2025a), knowledge distillation strategies (Le et al., 2025; Zhao et al., 2024), and transfer learning strategies (Yi et al., 2024). In addition, graph-based FL methods have been increasingly investigated to explicitly model structural dependencies among clients (Dinh et al., 2022; Sattler et al., 2021). These methods typically encode inter-client relationships from model parameters, local updates, gradients, or behavioral representations (Ge et al., 2024). The inferred topology can then be used to estimate client importance, guide model aggregation, or facilitate structured knowledge propagation across clients (Abed et al., 2026). As a representative category of graph-based FL, GCN-based methods model clients as graph nodes and employ trainable graph-convolutional layers to aggregate information from graph neighborhoods (Song et al., 2025). Although these methods provide an effective means of exploiting inter-client structure, their aggregation behavior is often jointly determined by the estimated topology and learned graph-convolution parameters. Moreover, client graphs are commonly constructed from predefined structures or instantaneous

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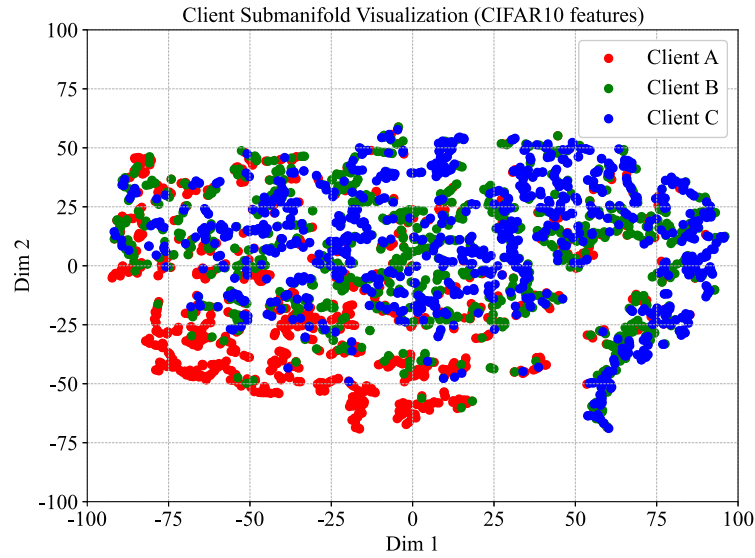


Fig. 1. Visualization of client-specific feature submanifolds.

observations at each communication round, which may render the resulting topology sensitive to transient update fluctuations and unable to capture the temporal evolution of inter-client dependencies.

Many aggregation-based FL methods implicitly assume that local model updates are sufficiently compatible within a shared parameterization, such that they can be directly combined to obtain an effective global model. Under heterogeneous data distributions, clients optimize different local objectives and may follow divergent optimization trajectories toward distinct local optima. As illustrated in Fig. 1, the feature representations extracted from Clients A, B, and C evolve along different trajectories and occupy distinguishable regions in the low-dimensional latent space. Weighted averaging may aggregate local updates that are geometrically misaligned and directionally conflicting, thereby preventing the global model from adequately reflecting the representational structure of any individual client. Moreover, the relationships among clients are not static, as the similarities among local model parameters, update gradients, and learned representations continuously evolve over successive communication rounds. A predefined client graph fails to capture the evolving nature of inter-client collaborative relationships, while a graph construction merely from the observations of a single round is overly responsive to momentary fluctuation in updates and the noise present in local training. Given that the inferred topology exerts a critical influence on the subsequent dissemination of information and model aggregation, this instability is highly likely to induce inconsistency in the collaborative schemes adopted over successive communication rounds. Therefore, the central challenge is to construct a sparse and adaptive collaboration topology that accommodates round-wise relationship evolution, and remains robust to transient local fluctuations.

Motivated by this, we propose Adaptive Topology-aware Federated Aggregation (ATFA), which organizes inter-client topology construction and topology-aware knowledge diffusion as two consecutive stages of the aggregation process. At each communication round, ATFA first calculates inter-client dependencies by incorporating historical trend information and then constructs a graph topology. Based on the resulting topology, ATFA subsequently performs structured knowledge fusion. The primary contributions are summarized as follows:

- We develop a round-adaptive sparse topology construction that integrates historical trend information with a density-calibrated threshold to capture persistent dependencies and affinities among clients.
- We design a Laplacian-based heat diffusion that performs topology-aware knowledge fusion over the adaptively constructed inter-client graph to enhance aggregation stability.

- We analyze the stabilization behavior of the evolving client topology and extensively evaluate ATFA on standard and medical imaging benchmark datasets. The experimental results demonstrate stable convergence and statistically significant overall improvements over representative baselines, supporting its robustness across the evaluated settings and its generalization across datasets.

The remainder of this paper is organized as follows. Section 2 reviews the related work with a focus on FL, aggregation strategies, and graph learning. Section 3 introduces the proposed ATFA method and offers a detailed description of its core components. Section 4 outlines the experimental setup and subsequently discusses the experimental results. Finally, Section 5 provides a comprehensive summary and highlights potential directions for future research.

2. Related work

2.1. Federated learning under data heterogeneity

Federated learning (FL) enables multiple clients to collaboratively train a shared model without directly exchanging their raw data. Nevertheless, conventional FL algorithms may suffer from slow convergence and degraded performance when client data follow heterogeneous non-IID distributions. A common line of research addresses this problem by modifying local objectives to constrain client drift and improve the compatibility of local updates with the global optimization (He et al., 2024). For instance, FedProx (Li et al., 2020) extended FedAvg by incorporating a proximal term that penalized excessive deviation of a local model from the current global model. Building on this concept, Acar et al. (2021) introduced a dynamically updated linear regularization term to align the stationary points of the local and global objectives, thereby reducing the inconsistency between local and global optimization. Related approaches have also employed auxiliary local models or additional consistency losses to suppress oscillatory local updates while avoiding the exchange of sensitive intermediate information (Tang et al., 2022b). Another group of methods mitigates heterogeneity by regularizing learned representations. For example, MOON (Li et al., 2021) introduced model-level contrastive learning, encouraging the representation generated by the current local model to remain close to that generated by the latest global model while separating it from the representation of the previous local model. Feature-level regularization methods similarly reduce representation shifts between local and global models (Pei, 2025), while other representation-oriented

approaches promote invariant distributions or statistical consistency across clients (Sun et al., 2024). Client drift has also been addressed through gradient correction and normalized aggregation. For example, SCAFFOLD (Karimireddy et al., 2020) employed server-side and client-side control variates to correct the local update directions induced by heterogeneous data. In contrast, FedNova (Wang et al., 2020) normalized local updates according to the effective number of local optimization steps, thereby correcting the objective inconsistency caused by heterogeneous local computation. Correlation-aware approaches, exemplified by FedCor (Tang et al., 2022a), explicitly capture the statistical correlations among client losses, thereby enabling more discriminative and informative client selection. Although existing regularization, contrastive learning, control-variate, and normalized-aggregation methods substantially mitigate client drift under non-IID data, most of them operate at the level of individual clients by constraining local optimization relative to the global model or independently correcting local updates. The structural dependencies among clients are not explicitly incorporated into the model aggregation process.

2.2. Model aggregation strategy

Server-side aggregation plays a critical role in determining the quality of the global model and the stability of federated optimization (Chen et al., 2021b; Du et al., 2024). In the classic FedAvg framework, model aggregation is performed as a weighted average of local models, with the weights determined by the amount of local data (Chen et al., 2019; Zhong et al., 2024). However, under strong non-IID data distributions, weighting strategies based on data volume may become suboptimal, updates associated with large local datasets may dominate the aggregation process even when they conflict with the global optimization direction, potentially leading to oscillatory training behavior and slower convergence (Niu et al., 2025). To alleviate this limitation, recent studies have developed adaptive aggregation strategies that assign client weights according to model similarity, update alignment, reliability, local performance, or other contribution-related indicators (Li et al., 2025). For example, Cheng et al. (2026) introduced FedHPLP, which constructed a learnable interaction graph via high- and low-pass filters and assigns aggregation weights based on each participant's sample redundancy. Fed-LAA (Khan et al., 2026) performed per-layer gradient alignment optimization to assign adaptive weights that reflected the contribution quality of each node-layer. Another line of research adopts clustered (Jiang et al., 2026), grouped (He et al., 2026), or hierarchical aggregation strategies (Ju et al., 2026), in which clients with relatively compatible updates are first aggregated within groups before their intermediate models are merged at a higher level (Luo et al., 2020; Wang et al., 2021). Despite these advances, most adaptive aggregation methods represent inter-client compatibility through independent scalar weights, pairwise scores, or discrete group assignments. These representations constrain the support for modeling higher-order dependencies and coordinated interactions among multiple clients.

2.3. Graph-based federated learning

Graph-based federated learning provides a structured approach to modeling dependencies and collaboration patterns among heterogeneous clients. One line of research constructs client similarity graphs to support clustering or groupwise aggregation. For example, Palihawadana et al. (2022) constructed a similarity graph from model information and used the resulting client groups to guide aggregation. Similarly, Yoo et al. (2022) introduced fuzzy clustering, allowing each client to possess soft memberships in multiple clusters and participate in different group-specific aggregation processes. Disjoint clustering methods instead partition clients into mutually exclusive groups, with the aim of reducing interference among incompatible clients and mitigating global model drift (Yu et al., 2024). Beyond clustering methods, relationship-aware methods use structured client dependencies to regulate informa-

tion exchange and facilitate knowledge transfer across heterogeneous local models. These methods may propagate model representations, prototypes, or other transferable knowledge along estimated client relations (Tan et al., 2022). Another line of research focuses on graph neural networks (GNNs), for enhancing aggregation or topology control. For example, Rizwan et al. (2026) proposed FedRGT, a coordinator-assisted framework that dynamically optimized network topology using a GNN. Sun et al. (2026) proposed FEDGE, a privacy-preserving heterogeneous GNN that leveraged local pseudo-node generation and federated graph enhancement to recover missing cross-client information, combined with an adaptive aggregation strategy guided by federated clustering factors. Despite these advances, existing graph-based FL methods still face several limitations in topology construction. Many approaches estimate graph edges from single-round pairwise similarities derived from model parameters, local updates, gradients, performance indicators, or learned representations. These instantaneous estimates may be sensitive to stochastic local optimization and temporary update deviations. Moreover, fixed thresholding or sparsification rules may fail to preserve an appropriate collaboration density when the distribution of inter-client similarities changes across communication rounds. In the GCN- and GNN-based methods, the effect of the inferred topology may be intertwined with the trainable message-passing parameters, making it difficult to decouple topology reliability from knowledge propagation.

As summarized in Table 1, some existing graph-based FL methods can be broadly categorized into graph-supported client clustering, relationship-aware aggregation or knowledge transfer. In particular, client relationships are often inferred from instantaneous observations, while graph sparsity is controlled through fixed neighborhoods or predefined thresholds. Furthermore, GNN-based approaches couple information propagation with trainable graph-convolutional parameters. In contrast, ATFA integrates topology stabilization, adaptive sparse graph construction, and structured knowledge diffusion within a unified aggregation framework. ATFA propagates local model information through a non-trainable heat kernel derived from the graph Laplacian. This design explicitly controls the reliability of the client topology before knowledge fusion and differs from trainable graph-convolutional propagation.

3. Adaptive topology-aware federated aggregation (ATFA)

This section offers a thorough overview of ATFA. ATFA dynamically constructs a client graph using historical trend information and performs topology-aware aggregation through heat diffusion. The framework of the proposed ATFA is illustrated in Fig. 2.

3.1. Problem formalization

We consider an FL system with N edge devices. Client i holds a private dataset $\mathcal{D}_i$, and $\mathcal{X} = \{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_N\}$ denotes the collection of client datasets. Each client i possesses a private local dataset $\mathcal{D}_i = \{(x_j^{(i)}, y_j^{(i)})\}_{j=1}^{n_i}$ of size n_i , performs local training on it, subsequently uploading the updated model $\theta_i^{(t)}$ to the server. The overall objective is to minimize the weighted loss aggregated from all local clients, which can be formulated as follows:

$$F(\theta^{(t)}) = \sum_{i=1}^N p_i F_i(\theta^{(t)}) = \sum_{i=1}^N p_i \left(\frac{1}{n_i} \sum_{(x_i, y_i) \in \mathcal{D}_i} \mathcal{L}(f(x_i; \theta^{(t)}), y_i) \right), \quad (1)$$

where $\theta^{(t)}$ denotes the global model parameters, $\mathcal{L}(\cdot)$ is the local loss function, and $f(x_i; \theta^{(t)})$ represents the prediction output of the model for input x_i . Here, the p_i denotes the aggregation weight. During each communication round, the server distributes the current global model parameters $\theta^{(t)}$ to the clients. Accordingly, global aggregation across

Table 1

Comparison of client-graph construction and aggregation mechanisms in representative graph-based FL methods.

Method	Model-Affinity Client Graph	Cross-Client Graph Convolution	Gradient-Based Graph	Influential-Client Learning	Historical Topology Stabilization	Density-Calibrated Sparsification	Aggregation Strategy
FedAGA (Ge et al., 2024)	✗	✗	✓	✗	✗	✗	Adaptive Graph-based Aggregation
TFL (Ma et al., 2024a)	✓	✗	✗	✗	✓	✗	Topology-aware Influence-weighted Aggregation
FedGraph (Chen et al., 2021a)	✗	✓	✗	✗	✗	✗	Mini-batch-size-weighted FedAvg
GCFL+ (Xie et al., 2021)	✗	✗	✓	✗	✗	✗	Gradient-based Cluster-wise Aggregation
SFL (Chen et al., 2022)	✓	✓	✗	✗	✗	✗	GCN-based client-wise aggregation
FedSKA (Li et al., 2023)	✗	✓	✗	✗	✗	✗	Dynamic GCN-based client representation aggregation
FedSim (Palihawadana et al., 2022)	✗	✗	✓	✓	✓	✓	Gradient-similarity-guided local-global aggregation
FedAGHN (Song et al., 2025)	✓	✓	✗	✗	✗	✗	Layer-wise attentive graph-hypernetwork aggregation
FedLDP (Yu et al., 2026)	✓	✗	✗	✗	✗	✗	Collaboration-graph-based personalized logits-prototype aggregation
ATFA (Ours)	✗	✗	✓	✗	✓	✓	Heat Diffusion Aggregation

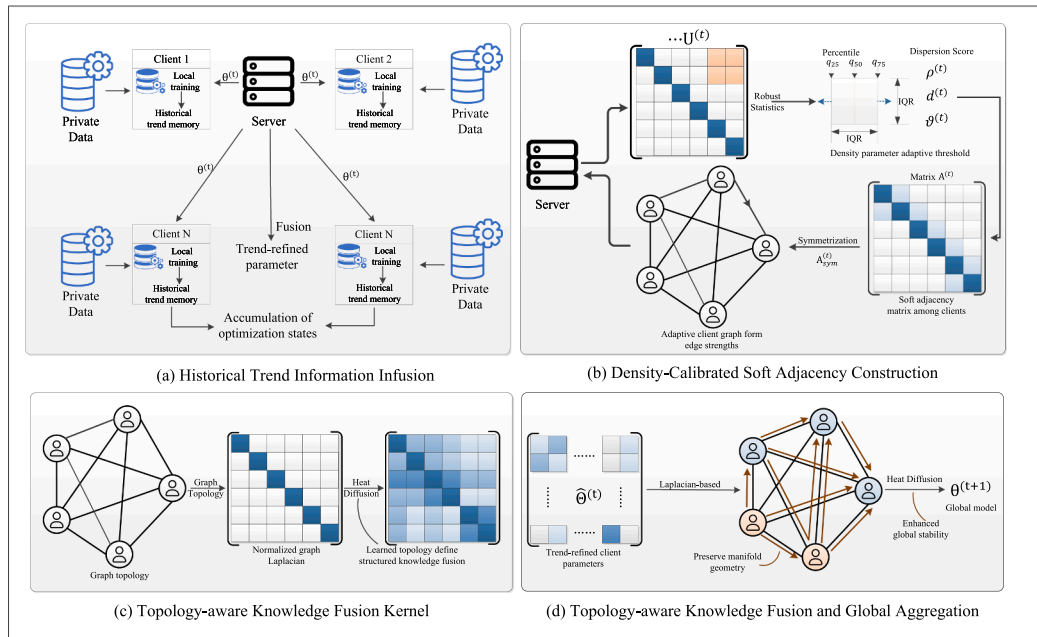


Fig. 2. Overview of ATFA. (a) The server collects local models and constructs historical trend information. (b) A dynamic graph topology is constructed to derive the corresponding Laplacian matrix. (c) A knowledge fusion kernel is formulated based on the graph topology. (d) Local models are subsequently aggregated into the global model based on topology-aware knowledge fusion with heat diffusion.

clients is performed as follows:

$$\theta^{(t+1)} = \sum_i p_i \theta_i^{(t)}, \quad p_i = \frac{n_i}{\sum_k n_k}. \quad (2)$$

3.2. Historical trend information infusion

Inter-client similarity derived from a single-round update may be unreliable due to stochastic local optimization. To address this issue, we maintain a historical trend information $s_i^{(t)} \in \mathbb{R}^d$ for each client i that accumulates the update deviation between the local models and the broadcasts global model. After completing local training in each round t , each client i updates its trend information as follows:

$$s_i^{(t)} = s_i^{(t-1)} + (\theta_i^{(t)} - \theta^{(t)}), \quad s_i^{(1)} = \mathbf{0}. \quad (3)$$

This allows the trend memory to capture each client's optimization-drift direction while retaining historical local information. To construct

a stable topology, we compute a trend-refined parameter vector $\hat{\theta}_i^{(t)}$ by integrating the current update with historical trend information, i.e.,

$$\hat{\theta}_i^{(t)} = (1 - \lambda)\theta_i^{(t)} + \lambda s_i^{(t)}, \quad \lambda \in [0, 1], \quad (4)$$

where infusion coefficient λ is consistently defined on $[0, 1]$, which controls the trade-off between short-term adaptation and long-term optimization consistency. This design can suppress high-frequency fluctuations arising from stochastic optimization.

3.3. Density-calibrated soft adjacency construction

For any pair of participating clients (i, j) with $i \neq j$, we compute the inter-client affinity using the cosine similarity between the historical

trend vectors, i.e.,

$$u_{ij}^{(t)} = \frac{\langle s_i^{(t)}, s_j^{(t)} \rangle}{\|s_i^{(t)}\|_2 \|s_j^{(t)}\|_2}, \quad i \neq j, \quad (5)$$

where $u_{ij}^{(t)} \in [-1, 1]$, and $\langle \cdot \rangle$ denotes the Euclidean inner product. The affinity matrix is denoted by $\mathbf{U}^{(t)} \in \mathbb{R}^{N \times N}$. To avoid trivial self-similarity, we collect all off-diagonal similarities, i.e.,

$$\Omega^{(t)} = \{u_{ij}^{(t)}, i \neq j\}, \quad |\Omega^{(t)}| = N(N-1). \quad (6)$$

We compute the 25th (q_{25}), 50th (q_{50}), and 75th (q_{75}) percentiles of $\Omega^{(t)}$,

$$(q_{25}, q_{50}, q_{75}) = \text{Percentile}(\Omega^{(t)}; 25, 50, 75). \quad (7)$$

To quantify the variability of the pairwise affinity set $\Omega^{(t)}$, we define a robust dispersion statistic that resists the influence of extreme similarities and distributional skewness. Specifically, we compute the interquartile range (IQR) and the mean absolute deviation around the median (MAD_{ev}) as follows:

$$\text{IQR}^{(t)} = (q_{75} - q_{25}) + \epsilon, \quad \text{MAD}_{ev}^{(t)} = \frac{1}{|\Omega^{(t)}|} \sum_{x \in \Omega^{(t)}} |x - q_{50}|. \quad (8)$$

Here, $\epsilon > 0$ is a small regularization constant introduced to prevent division by zero. Based on these robust statistics, we construct a dimensionless dispersion score, i.e.,

$$\rho^{(t)} = \frac{\text{MAD}_{ev}^{(t)}}{\text{IQR}^{(t)}}. \quad (9)$$

This score is mapped to a bounded density parameter $d^{(t)} \in (0, 1)$ via a sigmoid function, i.e.,

$$d^{(t)} = \frac{1}{1 + e^{(1-\rho^{(t)})}}. \quad (10)$$

The resulting density parameter controls the proportion of pairwise affinities retained in the client graph. Accordingly, the adaptive threshold $\tau^{(t)}$ is defined as an empirical quantile of $\Omega^{(t)}$, i.e.,

$$\tau^{(t)} = Q_{\Omega^{(t)}}(1 - d^{(t)}), \quad (11)$$

where $Q_{\Omega}(\cdot)$ denotes the empirical quantile function. Based on this threshold, we construct a soft adjacency matrix $\mathbf{A}^{(t)} \in [0, 1]^{N \times N}$, i.e.,

$$A_{ij}^{(t)} = \sigma \left(\frac{u_{ij}^{(t)} - \tau^{(t)}}{\text{IQR}^{(t)}} \right), \quad i \neq j, \quad (12)$$

and set $A_{ii}^{(t)} = 0$. Finally, we symmetrize the adjacency matrix as $\mathbf{A}_{\text{sym}}^{(t)} \triangleq \frac{1}{2}(\mathbf{A}^{(t)} + (\mathbf{A}^{(t)})^\top)$.

3.4. Topology-aware knowledge fusion

To achieve robust knowledge fusion across heterogeneous clients, we stack the trend-refined parameter vector of each client $\hat{\theta}_i^{(t)} \in \mathbb{R}^d$ into $\hat{\Theta}^{(t)} \in \mathbb{R}^{N \times d}$, i.e.,

$$\hat{\Theta}^{(t)} \triangleq \begin{bmatrix} (\hat{\theta}_1^{(t)})^\top \\ (\hat{\theta}_2^{(t)})^\top \\ \vdots \\ (\hat{\theta}_{N^{(t)}}^{(t)})^\top \end{bmatrix} \in \mathbb{R}^{N^{(t)} \times d}. \quad (13)$$

Rather than treating client updates as independent entities, we perform topology-aware knowledge fusion by diffusing client model information over the constructed collaboration graph. We adopt the symmetric normalized graph Laplacian, which compensates for degree variations and provides a degree-balanced and numerically stable basis for heat diffusion. It is defined as

$$L^{(t)} = I - D^{(t)^{-\frac{1}{2}}} \mathbf{A}_{\text{sym}}^{(t)} D^{(t)^{-\frac{1}{2}}}. \quad (14)$$

where $D^{(t)}$ is the degree matrix of $\mathbf{A}_{\text{sym}}^{(t)}$. The diffused client-parameter matrix is then computed using the Laplacian-based heat kernel as follows:

$$\tilde{\Theta}^{(t)} = \mathbf{H}^{(t)} \hat{\Theta}^{(t)} = \exp(-\gamma L^{(t)}) \hat{\Theta}^{(t)}, \gamma > 0 \quad (15)$$

where $\mathbf{H}^{(t)}$ is the heat-diffusion kernel defined as $\mathbf{H}^{(t)} = \exp(-\gamma L^{(t)})$. The parameter γ controls the effective diffusion time on the learned client graph (Gasteiger et al., 2019). A small γ leads to a confined and spatially localized information exchange, which in turn preserves the majority of client-specific information. Increasing γ extends the effective propagation scale and strengthens knowledge fusion among structurally related clients.

The next-round global model is obtained by aggregating the heat diffusion-based client models as follows:

$$\theta^{(t+1)} = \frac{p^\top \tilde{\Theta}^{(t)}}{p^\top \mathbf{H}^{(t)} \mathbf{1}_N} = \frac{p^\top \mathbf{H}^{(t)} \hat{\Theta}^{(t)}}{p^\top \mathbf{H}^{(t)} \mathbf{1}_N}, \quad (16)$$

where $\mathbf{1}_N \in \mathbb{R}^N$ denotes the all-ones vector. The denominator normalizes the topology-induced aggregation weights. For clarity, define $Z_t = p^\top \mathbf{H}^{(t)} \mathbf{1}_N$, and $(q^{(t)})^\top = \frac{p^\top \mathbf{H}^{(t)}}{Z_t}$. The aggregation strategy can then be equivalently expressed as

$$\theta^{(t+1)} = \sum_{i=1}^N q_i^{(t)} \hat{\theta}_i^{(t)}. \quad (17)$$

Since $-L^{(t)}$ is a Metzler matrix, the heat kernel is entrywise nonnegative. Moreover, because $p_i > 0$, we have $Z_t > 0$, $q_i^{(t)} > 0$, and $\sum_{i=1}^N q_i^{(t)} = 1$.

Therefore, the proposed aggregation strategy produces a valid convex combination of the trend-refined client models, while allowing the learned client topology to adaptively regulate the effective contribution of each client. ATFA performs heat diffusion over the graph topology according to the learned graph structure, thereby facilitating knowledge fusion among strongly coupled clients in the parameter space and ultimately obtaining the robustness of the global model. Algorithm 1 provides a detailed description of ATFA.

Algorithm 1 Adaptive topology-aware federated aggregation (ATFA).

- 1: **Input:** Communication rounds T , local epochs E , local learning rate η , infusion coefficient λ , heat diffusion coefficient γ , client dataset $\mathcal{X} = \{D_1, \dots, D_N\}$.
 - 2: **Output:** Global model $\theta^{(T)}$.
 - 3: Initialize global model and historical trend information;
 - 4: **for** $t = 1$ to T **do**
 - 5: **for** each client i **in parallel do**
 - 6: Receive global model $\theta^{(t)}$ and initialize local models;
 - 7: **for** $e = 1$ to E **do**
 - 8: **for** mini-batch $(x, y) \in D_i$ **do**
 - 9: Update local model by SGD;
 - 10: **end for**
 - 11: **end for**
 - 12: Update historical trend information.
 - 13: **end for**
 - 14: Construct pairwise affinity set over clients using cosine similarity;
 - 15: Collect off-diagonal similarities $\Omega^{(t)}$;
 - 16: Compute robust statistics (q_{25}, q_{50}, q_{75}), $\text{IQR}^{(t)}$, and $\text{MAD}_{ev}^{(t)}$ from $\Omega^{(t)}$;
 - 17: Determine adaptive threshold $\tau^{(t)}$;
 - 18: Construct soft adjacency matrix $\mathbf{A}^{(t)}$ and symmetrize it, denoted as $\mathbf{A}_{\text{sym}}^{(t)}$;
 - 19: Stack refined client parameters $\hat{\Theta}^{(t)}$;
 - 20: Compute degree matrix and normalized Laplacian;
 - 21: Aggregate the heat diffusion-based client models.
 - 22: **end for**
 - 23: **return** Global model $\theta^{(T)}$.
-

3.5. Theoretical analysis

This section provides a theoretical analysis of ATFA. The convergence analysis is stated for full client participation. Under partial participation, an additional client-sampling error term must be included in the centered stochastic error defined below.

Assumption 1 (Smoothness and strong convexity). Each local objective function $F_i : \mathbb{R}^d \rightarrow \mathbb{R}$ is L -smooth and μ -strongly convex. Thus, for all $v, w \in \mathbb{R}^d$,

$$\|\nabla F_i(v) - \nabla F_i(w)\| \leq L\|v - w\|,$$

and

$$F_i(v) \geq F_i(w) + \langle \nabla F_i(w), v - w \rangle + \frac{\mu}{2}\|v - w\|^2.$$

The global objective is

$$F(\theta) = \sum_{i=1}^N p_i F_i(\theta), \quad p_i > 0, \quad \sum_{i=1}^N p_i = 1,$$

and its unique minimizer is denoted by θ^* .

Assumption 2 (Unbiased stochastic gradients with bounded moments). For every iterate generated by the algorithm, the stochastic gradient $g_i(w, \xi)$ used by client i satisfies

$$\mathbb{E}[g_i(w, \xi) \mid w] = \nabla F_i(w),$$

$$\mathbb{E}\|g_i(w, \xi) - \nabla F_i(w)\|^2 \leq \sigma^2, \quad \mathbb{E}\|g_i(w, \xi)\|^2 \leq G^2.$$

Assumption 3 (Bounded heterogeneity). The degree of non-IIDness is quantified by the gradient dissimilarity,

$$\|\nabla F_i(\theta) - \nabla F(\theta)\| \leq \delta_{\text{het}}, \quad \forall \theta, i.$$

Remark. The bounded-second-moment condition in [Assumption 2](#) is required only along the iterates generated by ATFA. This avoids imposing a globally bounded gradient over the entire space $\mathbb{R}^d$, which would generally be incompatible with global strong convexity.

3.5.1. Global model convergence

Let F_t denote the history available immediately before the local updates in communication round t , and let $\mathbb{E}_t[\cdot] = \mathbb{E}[\cdot \mid F_t]$. Client i initializes $w_{i,0}^{(t)} = \theta^{(t)}$ and performs E local SGD steps:

$$w_{i,e+1}^{(t)} = w_{i,e}^{(t)} - \eta_t g_{i,e}^{(t)}, \quad e = 0, \dots, E-1,$$

where $g_{i,e}^{(t)} = g_i(w_{i,e}^{(t)}, \xi_{i,e}^{(t)})$. Define

$$G_i^{(t)} = \frac{1}{E} \sum_{e=0}^{E-1} g_{i,e}^{(t)}, \quad \theta_i^{(t)} = w_{i,E}^{(t)}, \quad \Delta_i^{(t)} = \theta_i^{(t)} - \theta^{(t)},$$

which implies

$$\Delta_i^{(t)} = -E\eta_t G_i^{(t)}.$$

The trend-refined local model admits the following exact equivalent form

$$\begin{aligned} \hat{\theta}_i^{(t)} &= (1 - \lambda_t)\theta_i^{(t)} + \lambda_t s_i^{(t)} \\ &= (1 - \lambda_t)(\theta^{(t)} + \Delta_i^{(t)}) + \lambda_t(s_i^{(t-1)} + \Delta_i^{(t)}) \\ &= \theta^{(t)} - E\eta_t G_i^{(t)} + \lambda_t(s_i^{(t-1)} - \theta^{(t)}). \end{aligned}$$

For the analysis, we allow a round-dependent infusion coefficient $\lambda_t \in [0, 1]$, and the fixed coefficient λ used in the method and experiments is the special case $\lambda_t \equiv \lambda$.

Lemma 1 (Local-update displacement). Under [Assumption 2](#), for every client i and local step e ,

$$\mathbb{E}_t\|w_{i,e}^{(t)} - \theta^{(t)}\|^2 \leq e^2 \eta_t^2 G^2.$$

In particular,

$$\mathbb{E}_t\|\Delta_i^{(t)}\|^2 \leq E^2 \eta_t^2 G^2.$$

Proof. The local recursion yields

$$w_{i,e}^{(t)} - \theta^{(t)} = -\eta_t \sum_{r=0}^{e-1} g_{i,r}^{(t)}.$$

Using $\|\sum_{r=0}^{e-1} x_r\|^2 \leq e \sum_{r=0}^{e-1} \|x_r\|^2$ and [Assumption 2](#),

$$\mathbb{E}_t\|w_{i,e}^{(t)} - \theta^{(t)}\|^2 \leq \eta_t^2 e \sum_{r=0}^{e-1} \mathbb{E}_t\|g_{i,r}^{(t)}\|^2 \leq e^2 \eta_t^2 G^2.$$

Setting $e = E$ gives the second claim. $\square$

Substituting the normalized ATFA aggregation rule above gives the exact recursion,

$$\begin{aligned} \theta^{(t+1)} &= \theta^{(t)} - E\eta_t \sum_{i=1}^N q_i^{(t)} G_i^{(t)} + \lambda_t \left(\sum_{i=1}^N q_i^{(t)} s_i^{(t-1)} - \theta^{(t)} \right) \\ &= \theta^{(t)} - \alpha_t G_t, \end{aligned}$$

where $\alpha_t = E\eta_t$ and $G_t = \sum_{i=1}^N q_i^{(t)} G_i^{(t)} + \frac{\lambda_t}{\alpha_t} \left(\theta^{(t)} - \sum_{i=1}^N q_i^{(t)} s_i^{(t-1)} \right)$.

Define

$$b_t = \mathbb{E}_t[G_t] - \nabla F(\theta^{(t)}), \quad \zeta_t = G_t - \mathbb{E}_t[G_t],$$

then $\mathbb{E}_t[\zeta_t] = 0$ and

$$\theta^{(t+1)} = \theta^{(t)} - \alpha_t (\nabla F(\theta^{(t)}) + b_t + \zeta_t).$$

The term b_t collects all ATFA-specific deviations from an unbiased stochastic gradient of the original global objective. Define

$$\chi_t = \mathbb{E}_t\|q^{(t)} - p\|_1,$$

$$a_t = \left\| \mathbb{E}_t \left[\frac{1}{E} \sum_{i=1}^N q_i^{(t)} \sum_{e=0}^{E-1} (g_{i,e}^{(t)} - \nabla F_i(w_{i,e}^{(t)})) \right] \right\|,$$

and

$$R_t = \left\| \mathbb{E}_t \left[\theta^{(t)} - \sum_{i=1}^N q_i^{(t)} s_i^{(t-1)} \right] \right\|.$$

Here χ_t measures the mismatch between the topology-induced effective weights and the target data weights, a_t captures possible correlation between the current stochastic-gradient noise and the graph constructed from the same-round updates, and R_t is the method-consistent residual between the broadcast model and the diffusion-weighted historical trend.

Proposition 1 (Bias Bound for the Gradient Estimator). Under [Assumptions 1–3](#),

$$\|b_t\| \leq B_t,$$

where

$$B_t = \frac{L\sqrt{N}(E-1)}{2} \eta_t G + \delta_{\text{het}} \chi_t + a_t + \frac{\lambda_t}{\alpha_t} R_t.$$

Proof. Add and subtract $\nabla F_i(\theta^{(t)})$ and the data-weighted global gradient. For the local-drift term, Jensen's inequality for the convex combination $q^{(t)}$ and [Lemma 1](#) give

$$\begin{aligned} \mathbb{E}_t \left\| \frac{1}{E} \sum_{i=1}^N q_i^{(t)} \sum_{e=0}^{E-1} (\nabla F_i(w_{i,e}^{(t)}) - \nabla F_i(\theta^{(t)})) \right\| \\ \leq \frac{L}{E} \sum_{e=0}^{E-1} \left(\sum_{i=1}^N \mathbb{E}_t \|w_{i,e}^{(t)} - \theta^{(t)}\|^2 \right)^{1/2} \\ \leq \frac{L\sqrt{N}}{E} \sum_{e=0}^{E-1} e \eta_t G = \frac{L\sqrt{N}(E-1)}{2} \eta_t G. \end{aligned}$$

Because $\sum_i (q_i^{(t)} - p_i) = 0$,

$$\left\| \sum_{i=1}^N (q_i^{(t)} - p_i) \nabla F_i(\theta^{(t)}) \right\| = \left\| \sum_{i=1}^N (q_i^{(t)} - p_i) (\nabla F_i(\theta^{(t)}) - \nabla F(\theta^{(t)})) \right\|$$

$$\leq \delta_{\text{het}} \|q^{(t)} - p\|_1.$$

Taking conditional expectation gives $\delta_{\text{het}} \chi_t$. The same-round noise-correlation term is exactly bounded by a_t , and the trend term is bounded by $(\lambda_t/\alpha_t)R_t$. Combining the four terms establishes the stated bias bound. $\square$

Lemma 2 (Deterministic Contraction with Summable Perturbations). *Let $x_t \geq 0$ satisfy*

$$x_{t+1} \leq (1 - c\alpha_t)x_t + \varepsilon_t,$$

where $c > 0$, $\varepsilon_t \geq 0$, $0 < c\alpha_t < 1$ for all sufficiently large t , $\sum_t \alpha_t = \infty$, and $\sum_t \varepsilon_t < \infty$. Then $x_t \rightarrow 0$.

Proof. Choose T_0 such that $0 < c\alpha_t < 1$ for all $t \geq T_0$. Unrolling the recursion from T_0 gives

$$x_{t+1} \leq \left(\prod_{r=0}^t (1 - c\alpha_r) \right) x_0 + \sum_{s=0}^t \left(\prod_{r=s+1}^t (1 - c\alpha_r) \right) \varepsilon_s.$$

Since $\sum_{r=T_0}^{\infty} \alpha_r = \infty$, the first product tends to zero. For the second term, fix $S \geq T_0$. Every term with $s \leq S$ tends to zero, whereas the remaining tail is bounded by $\sum_{s>S} \varepsilon_s$, which can be made arbitrarily small. Hence $x_t \rightarrow 0$. $\square$

Theorem 1 (Conditional Global-Model Convergence under Summable ATFA Bias). *Suppose Assumptions 1–3, and the normalized ATFA aggregation rule defined above hold, with the bias bounded as in Proposition 1. Assume additionally that the centered estimator has uniformly bounded conditional variance:*

$$\mathbb{E}_t \|\zeta_t\|^2 \leq V < \infty.$$

Let the deterministic effective learning rate $\alpha_t = E\eta_t$ satisfy

$$\sum_{t=0}^{\infty} \alpha_t = \infty, \quad \sum_{t=0}^{\infty} \alpha_t^2 < \infty, \quad 0 < \alpha_t \leq \frac{\mu}{4L^2}.$$

If

$$\sum_{t=0}^{\infty} \alpha_t \mathbb{E}[B_t^2] < \infty,$$

then

$$\lim_{t \rightarrow \infty} \mathbb{E} \|\theta^{(t)} - \theta^*\|^2 = 0,$$

and

$$\lim_{t \rightarrow \infty} \mathbb{E} [F(\theta^{(t)}) - F(\theta^*)] = 0.$$

Moreover, if $\alpha_t = a/(t + t_0)$ with $a > 2/\mu$, t_0 sufficiently large, and

$$\mathbb{E}[B_t^2] \leq C_B \alpha_t,$$

then

$$\mathbb{E} \|\theta^{(t)} - \theta^*\|^2 = \mathcal{O}(1/t), \quad \mathbb{E} [F(\theta^{(t)}) - F(\theta^*)] = \mathcal{O}(1/t).$$

Proof. Let $e_t = \theta^{(t)} - \theta^*$. From the exact global recursion,

$$e_{t+1} = e_t - \alpha_t (\nabla F(\theta^{(t)}) + b_t + \zeta_t).$$

Taking conditional expectation and using $\mathbb{E}_t[\zeta_t] = 0$,

$$\begin{aligned} \mathbb{E}_t \|e_{t+1}\|^2 &= \left\| e_t - \alpha_t (\nabla F(\theta^{(t)}) + b_t) \right\|^2 + \alpha_t^2 \mathbb{E}_t \|\zeta_t\|^2 \\ &\leq \|e_t\|^2 - 2\alpha_t \langle e_t, \nabla F(\theta^{(t)}) \rangle + 2\alpha_t \|e_t\| \|b_t\| \\ &\quad + 2\alpha_t^2 \|\nabla F(\theta^{(t)})\|^2 + 2\alpha_t^2 \|b_t\|^2 + \alpha_t^2 V. \end{aligned}$$

Strong convexity and smoothness imply,

$$\langle e_t, \nabla F(\theta^{(t)}) \rangle \geq \mu \|e_t\|^2, \quad \|\nabla F(\theta^{(t)})\| \leq L \|e_t\|.$$

Young's inequality gives,

$$2\alpha_t \|e_t\| \|b_t\| \leq \mu \alpha_t \|e_t\|^2 + \frac{\alpha_t}{\mu} \|b_t\|^2,$$

consequently,

$$\begin{aligned} \mathbb{E}_t \|e_{t+1}\|^2 &\leq (1 - \mu \alpha_t + 2L^2 \alpha_t^2) \|e_t\|^2 + \left(\frac{\alpha_t}{\mu} + 2\alpha_t^2 \right) B_t^2 + \alpha_t^2 V \\ &\leq \left(1 - \frac{\mu}{2} \alpha_t \right) \|e_t\|^2 + \left(\frac{\alpha_t}{\mu} + 2\alpha_t^2 \right) B_t^2 + \alpha_t^2 V, \end{aligned}$$

where the final inequality uses $\alpha_t \leq \mu/(4L^2)$. Taking total expectation and setting $x_t = \mathbb{E} \|e_t\|^2$ gives

$$x_{t+1} \leq \left(1 - \frac{\mu}{2} \alpha_t \right) x_t + \left(\frac{\alpha_t}{\mu} + 2\alpha_t^2 \right) \mathbb{E}[B_t^2] + \alpha_t^2 V.$$

The learning-rate conditions and the bias-summability condition above imply that the perturbation sequence is summable. Lemma 2 therefore yields $\mathbb{E} \|\theta^{(t)} - \theta^*\|^2 \rightarrow 0$. Since F is L -smooth and $\nabla F(\theta^*) = 0$,

$$0 \leq F(\theta^{(t)}) - F(\theta^*) \leq \frac{L}{2} \|\theta^{(t)} - \theta^*\|^2,$$

which proves convergence of the expected objective gap.

For the rate result, $\mathbb{E}[B_t^2] \leq C_B \alpha_t$ implies that the additive term is at most $C \alpha_t^2$ for a finite constant C after taking total expectation. Therefore,

$$x_{t+1} \leq \left(1 - \frac{\mu \alpha}{2(t + t_0)} \right) x_t + \frac{C'}{(t + t_0)^2}.$$

Because $\frac{\mu \alpha}{2} > 1$, a standard induction argument gives $x_t = \mathcal{O}(1/t)$. The objective-gap rate follows from smoothness. $\square$

Corollary 1 (Fixed-step Convergence to a Neighborhood). *Suppose the conditions of Proposition 1 hold with a constant step size $\alpha \leq \mu/(4L^2)$, $\sup_t \mathbb{E}[B_t^2] \leq B^2$, and $\mathbb{E}_t \|\zeta_t\|^2 \leq V$. Then*

$$\mathbb{E} \|\theta^{(t)} - \theta^*\|^2 \leq \left(1 - \frac{\mu \alpha}{2} \right)^t \|\theta^{(0)} - \theta^*\|^2 + \frac{2B^2}{\mu^2} + \frac{4\alpha B^2}{\mu} + \frac{2\alpha V}{\mu}.$$

Thus, with the constant learning rate used in the experiments, the global model contracts geometrically toward a stable neighborhood. The neighborhood size is controlled by local-update drift, topology-induced weight mismatch, same-round adaptive-weight correlation, the historical-trend residual, and stochastic-gradient variance.

Proof. Let $e_t = \theta^{(t)} - \theta^*$ as above. Taking total expectation gives

$$\mathbb{E} \|e_{t+1}\|^2 \leq \left(1 - \frac{\mu \alpha}{2} \right) \mathbb{E} \|e_t\|^2 + \frac{\alpha B^2}{\mu} + 2\alpha^2 B^2 + \alpha^2 V,$$

unrolling the recursion and using

$$\sum_{r=0}^{t-1} \left(1 - \frac{\mu \alpha}{2} \right)^r \leq \frac{2}{\mu \alpha}$$

gives the stated bound. $\square$

Remark. The bias-summability condition above explicitly identifies the quantities that must be controlled. For example, a one-round-delayed graph makes $q^{(t)}$ measurable with respect to $\mathcal{F}_t$ and therefore gives $a_t = 0$. For an optional centered-residual variant, define

$$\bar{r}_i^{(t)} = (s_i^{(t-1)} - \theta^{(t)}) - \sum_j q_j^{(t)} (s_j^{(t-1)} - \theta^{(t)}) = s_i^{(t-1)} - \sum_j q_j^{(t)} s_j^{(t-1)}$$

which satisfies $\sum_i q_i^{(t)} \bar{r}_i^{(t)} = 0$. This centering changes the injected residual and is not automatically satisfied by the current cumulative-trend definition. More generally, the rate condition $\mathbb{E}[B_t^2] = \mathcal{O}(\alpha_t)$ follows if the weight mismatch, same-round correlation, and the scaled corrected residual $\frac{\lambda_t}{\alpha_t} R_t$ are all $\mathcal{O}(\sqrt{\alpha_t})$ in mean square. A fixed, non-vanishing residual $\theta^{(t)} - \sum_i q_i^{(t)} s_i^{(t-1)}$ produces a persistent optimization bias and cannot be ignored.

3.5.2. Asymptotic regularity of the learned topology

The following analysis establishes that the round-to-round topology variation vanishes. This property is weaker than convergence to a fixed adjacency matrix, the distinction is stated explicitly below. Assume that the trend state satisfies the update relation

$$s_i^{(t)} - s_i^{(t-1)} = \theta_i^{(t)} - \theta_i^{(t-1)} = \Delta_i^{(t)}.$$

For $i \neq j$, define the cosine affinity

$$u_{ij}^{(t)} = \frac{\langle s_i^{(t)}, s_j^{(t)} \rangle}{\|s_i^{(t)}\|_2 \|s_j^{(t)}\|_2},$$

and set $u_{ii}^{(t)} = 0$. Let $U^{(t)} = [u_{ij}^{(t)}]$.

Assumption 4 (Non-degenerate trend states). There exists $\delta_s > 0$ such that, after a finite burn-in period,

$$\|s_i^{(t)}\|_2 \geq \delta_s$$

for every client i .

Proposition 2 (Vanishing Variation of Cosine Affinities). Under Assumption 4 and Lemma 1, for every pair $i \neq j$,

$$\mathbb{E}_t |u_{ij}^{(t)} - u_{ij}^{(t-1)}| \leq \frac{4EG}{\delta_s} \eta_t.$$

Let $M = N(N-1)$ be the number of off-diagonal entries. Then

$$\mathbb{E}_t \|U^{(t)} - U^{(t-1)}\|_F \leq \frac{4EG\sqrt{M}}{\delta_s} \eta_t,$$

and

$$\mathbb{E}_t \|U^{(t)} - U^{(t-1)}\|_{\max} \leq \frac{4EG\sqrt{M}}{\delta_s} \eta_t.$$

$\|X\|_{\max} := \max_{i,j} |X_{ij}|$ denotes the entrywise maximum norm. Consequently, if $\eta_t \rightarrow 0$, the affinity variation converges to zero in L^1 and hence in probability.

Proof. Define $n(x) = x/\|x\|_2$. If $\|x\|_2, \|y\|_2 \geq \delta_s$, then

$$\left\| \frac{x}{\|x\|_2} - \frac{y}{\|y\|_2} \right\|_2 \leq \frac{2}{\delta_s} \|x - y\|_2.$$

Since $u_{ij}^{(t)} = \langle n(s_i^{(t)}), n(s_j^{(t)}) \rangle$ and $\|n(x)\|_2 = 1$,

$$\begin{aligned} |u_{ij}^{(t)} - u_{ij}^{(t-1)}| &\leq \|n(s_i^{(t)}) - n(s_i^{(t-1)})\|_2 + \|n(s_j^{(t)}) - n(s_j^{(t-1)})\|_2 \\ &\leq \frac{2}{\delta_s} (\|s_i^{(t)}\|_2 + \|s_j^{(t)}\|_2), \end{aligned}$$

where the trend-state increment identity above was used. By Cauchy-Schwarz and Lemma 1,

$$\mathbb{E}_t \|\Delta_i^{(t)}\|_2 \leq \left(\mathbb{E}_t \|\Delta_i^{(t)}\|_2^2 \right)^{1/2} \leq E \eta_t G.$$

This gives the pairwise bound. Moreover,

$$\mathbb{E}_t |u_{ij}^{(t)} - u_{ij}^{(t-1)}|^2 \leq \frac{16E^2 \eta_t^2 G^2}{\delta_s^2},$$

summing over the M off-diagonal entries and applying Jensen's inequality yields

$$\mathbb{E}_t \|U^{(t)} - U^{(t-1)}\|_F \leq \frac{4EG\sqrt{M}}{\delta_s} \eta_t.$$

The entrywise max norm is bounded by the Frobenius norm, which proves the final inequality. $\square$

Let $\Omega^{(t)} = \{u_{ij}^{(t)} : i \neq j\}$ be the multiset of off-diagonal affinities, and let $Q_t(\beta)$ denote its linearly interpolated empirical quantile at level $\beta \in (0, 1)$. Define

$$\beta = 1 - d^{(t)} \tau^{(t)} = Q_t(\beta).$$

Assumption 5 (Regularity of the adaptive quantile level). There exists $\kappa_q > 0$ and an interval $I \subset (0, 1)$ containing all sufficiently late β such that

$$|Q_t(\beta) - Q_t(\beta')| \leq \kappa_q |\beta - \beta'|, \quad \forall \beta, \beta' \in I.$$

Lemma 3 (Stability of the Density Parameter). Let $\Delta_{\max,t} = \|U^{(t)} - U^{(t-1)}\|_{\max}$. For the density parameter $d^{(t)}$ defined through the robust dispersion score and sigmoid mapping in the method section,

$$|d^{(t)} - d^{(t-1)}| \leq C_d \Delta_{\max,t}, \quad C_d = \frac{1}{2\epsilon_{\text{IQR}}} + \frac{1}{\epsilon_{\text{IQR}}^2}.$$

Consequently, Proposition 2 implies $\mathbb{E}|d^{(t)} - d^{(t-1)}| = \mathcal{O}(\eta_t)$.

Proof. Fixed-level empirical quantiles are 1-Lipschitz under $\Delta_{\max,t}$. Hence the regularized IQR denominator and the mean absolute deviation around the median satisfy

$$|I_\epsilon^{(t)} - I_\epsilon^{(t-1)}| \leq 2\Delta_{\max,t}, \quad |\text{MAD}_{\text{ev}}^{(t)} - \text{MAD}_{\text{ev}}^{(t-1)}| \leq 2\Delta_{\max,t},$$

where $I_\epsilon^{(t)} = Q_t(0.75) - Q_t(0.25) + \epsilon_{\text{IQR}} \geq \epsilon_{\text{IQR}}$. Since $\text{MAD}_{\text{ev}}^{(t)} \leq 2$, the ratio $\rho^{(t)} = \text{MAD}_{\text{ev}}^{(t)} / I_\epsilon^{(t)}$ obeys

$$|\rho^{(t)} - \rho^{(t-1)}| \leq \left(\frac{2}{\epsilon_{\text{IQR}}} + \frac{4}{\epsilon_{\text{IQR}}^2} \right) \Delta_{\max,t}.$$

Finally, $d^{(t)} = \sigma(\rho^{(t)} - 1)$ and the logistic sigmoid is 1/4-Lipschitz, which proves the claim. $\square$

Remark. Assumption 5 is stated directly for the interpolated empirical quantile map rather than for the empirical CDF, which is a step function and does not possess an ordinary density. The assumption rules out abrupt changes in the selected order statistic when the target density level changes by a small amount.

Lemma 4 (Stability of Empirical Quantiles and IQR). Let

$$\Delta_{\max,t} = \|U^{(t)} - U^{(t-1)}\|_{\max}.$$

Under Assumption 5,

$$|\tau^{(t)} - \tau^{(t-1)}| \leq \Delta_{\max,t} + \kappa_q |d^{(t)} - d^{(t-1)}|.$$

Moreover, if

$$\text{IQR}^{(t)} = Q_t(0.75) - Q_t(0.25),$$

then

$$|\text{IQR}^{(t)} - \text{IQR}^{(t-1)}| \leq 2\Delta_{\max,t}.$$

Proof. If two finite samples have corresponding entries that differ by at most $\Delta_{\max,t}$, then every order statistic, and hence every interpolated empirical quantile at a fixed level, differs by at most $\Delta_{\max,t}$. Therefore,

$$|Q_t(\rho_{t-1}) - Q_{t-1}(\rho_{t-1})| \leq \Delta_{\max,t}.$$

Using the triangle inequality and Assumption 5,

$$\begin{aligned} |\tau^{(t)} - \tau^{(t-1)}| &\leq |Q_t(\rho_t) - Q_t(\rho_{t-1})| + |Q_t(\rho_{t-1}) - Q_{t-1}(\rho_{t-1})| \\ &\leq \kappa_q |\rho_t - \rho_{t-1}| + \Delta_{\max,t} \\ &= \kappa_q |d^{(t)} - d^{(t-1)}| + \Delta_{\max,t}. \end{aligned}$$

Applying the fixed-level quantile bound separately at levels 0.75 and 0.25 yields the stated IQR stability bound. $\square$

Define the regularized scale,

$$r_t = \text{IQR}^{(t)} + \epsilon_{\text{IQR}}, \quad \epsilon_{\text{IQR}} > 0.$$

Here $\text{IQR}^{(t)} = Q_t(0.75) - Q_t(0.25)$ is the unregularized spread, so r_t is the regularized IQR denominator used in the method section. We then construct the density-calibrated soft adjacency as

$$A_{ij}^{(t)} = \sigma \left(\frac{u_{ij}^{(t)} - \tau^{(t)}}{r_t} \right), \quad i \neq j, \quad A_{ii}^{(t)} = 0.$$

Because $u_{ij}^{(t)} = u_{ji}^{(t)}$ and $\tau^{(t)}$ and r_t are global scalars, $A^{(t)}$ is symmetric. Equivalently, one may use the symmetrized matrix specified in the method section.

Proposition 3 (Lipschitz Bound for the Soft Adjacency). *Let $M = N(N-1)$. Under [Assumption 5](#),*

$$\|A^{(t)} - A^{(t-1)}\|_F \leq c_F \|U^{(t)} - U^{(t-1)}\|_F + c_{\max} \|U^{(t)} - U^{(t-1)}\|_{\max} + c_d |d^{(t)} - d^{(t-1)}|, \quad (18)$$

where

$$c_F = \frac{1}{4\epsilon_{\text{IQR}}},$$

$$c_{\max} = \sqrt{M} \left(\frac{1}{4\epsilon_{\text{IQR}}} + \frac{1}{\epsilon_{\text{IQR}}^2} \right), \quad c_d = \frac{\sqrt{M}\kappa_q}{4\epsilon_{\text{IQR}}}.$$

Proof. Let $z_{ij}^{(t)} = \frac{u_{ij}^{(t)} - \tau^{(t)}}{r_t}$. The logistic sigmoid is globally Lipschitz with constant $1/4$, so

$$|A_{ij}^{(t)} - A_{ij}^{(t-1)}| \leq \frac{1}{4} |z_{ij}^{(t)} - z_{ij}^{(t-1)}|.$$

Because cosine affinities and their quantiles lie in $[-1, 1]$,

$$|u_{ij}^{(t-1)} - \tau^{(t-1)}| \leq 2.$$

Moreover, $r_t, r_{t-1} \geq \epsilon_{\text{IQR}}$, and [Lemma 4](#) gives

$$\left| \frac{1}{r_t} - \frac{1}{r_{t-1}} \right| = \frac{|\text{IQR}^{(t)} - \text{IQR}^{(t-1)}|}{r_t r_{t-1}} \leq \frac{2\Delta_{\max,t}}{\epsilon_{\text{IQR}}^2}.$$

Therefore,

$$|z_{ij}^{(t)} - z_{ij}^{(t-1)}| \leq \frac{|u_{ij}^{(t)} - u_{ij}^{(t-1)}|}{\epsilon_{\text{IQR}}} + \frac{|\tau^{(t)} - \tau^{(t-1)}|}{\epsilon_{\text{IQR}}} + \frac{4\Delta_{\max,t}}{\epsilon_{\text{IQR}}^2}.$$

Using [Lemma 4](#) and the triangle inequality for the Frobenius norm gives

$$\begin{aligned} \|A^{(t)} - A^{(t-1)}\|_F &\leq \frac{1}{4\epsilon_{\text{IQR}}} \|U^{(t)} - U^{(t-1)}\|_F \\ &\quad + \sqrt{M} \left(\frac{1}{4\epsilon_{\text{IQR}}} + \frac{1}{\epsilon_{\text{IQR}}^2} \right) \Delta_{\max,t} \\ &\quad + \frac{\sqrt{M}\kappa_q}{4\epsilon_{\text{IQR}}} |d^{(t)} - d^{(t-1)}|, \end{aligned}$$

which establishes the stated Lipschitz bound. $\square$

Theorem 2 (Asymptotic Regularity of the Learned Topology). *Suppose [Assumptions 4, 5](#), and the conditions of [Proposition 2](#) hold. If $\eta_t \rightarrow 0$, then*

$$\mathbb{E} \|A^{(t)} - A^{(t-1)}\|_F \rightarrow 0.$$

In particular, the round-to-round topology variation converges to zero in L^1 and hence in probability.

Proof. Taking expectations in the Lipschitz bound above and applying [Proposition 2](#) gives

$$\mathbb{E} \|A^{(t)} - A^{(t-1)}\|_F \leq \tilde{C}_A \eta_t,$$

where

$$\tilde{C}_A = \frac{4EG\sqrt{M}}{\delta_s} (c_F + c_{\max} + c_d C_d).$$

Since $\eta_t \rightarrow 0$, the right-hand side tends to zero, which proves the claim. $\square$

[Theorem 2](#) proves asymptotic regularity,

$$A^{(t)} - A^{(t-1)} \rightarrow 0,$$

in L^1 and in probability. This does not by itself imply the existence of a fixed limit A^∞ . A stronger sufficient condition is

$$\sum_{t=1}^{\infty} \eta_t < \infty,$$

By [Lemma 3](#), this single condition also yields $\sum_t \mathbb{E} |d^{(t)} - d^{(t-1)}| < \infty$, and therefore $\sum_t \mathbb{E} \|A^{(t)} - A^{(t-1)}\|_F < \infty$, making $\{A^{(t)}\}$ a Cauchy sequence in L^1 . This stronger condition is used only to guarantee convergence of the topology itself; with fixed E , it is not imposed in the global-model convergence theorem above because it would contradict $\sum_t \alpha_t = \infty$ when $\alpha_t = E\eta_t$.

3.5.3. Stability continuity heat diffusion

Proposition 4 (Non-Expansiveness of Heat Diffusion). *Let $\mathcal{L}^{(t)} \geq 0$ be a symmetric normalized graph Laplacian and*

$$\mathbf{H}^{(t)} = \exp(-\gamma \mathcal{L}^{(t)}), \quad \gamma > 0,$$

then

$$\|\mathbf{H}^{(t)}\|_2 \leq 1, \quad \|\mathbf{H}^{(t)} X\|_F \leq \|X\|_F, \quad \forall X.$$

Proof. Let

$$\mathcal{L}^{(t)} = Q \Lambda Q^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_N), \quad \lambda_k \geq 0,$$

then

$$\mathbf{H}^{(t)} = Q \text{diag}(e^{-\gamma \lambda_1}, \dots, e^{-\gamma \lambda_N}) Q^\top,$$

hence

$$\|\mathbf{H}^{(t)}\|_2 = \max_k e^{-\gamma \lambda_k} \leq 1.$$

The Frobenius-norm inequality follows from $\|AX\|_F \leq \|A\|_2 \|X\|_F$. $\square$

Proposition 5 (Temporal Continuity of the Heat Kernel). *Let*

$$\mathcal{L}(A) = I - D(A)^{-1/2} A D(A)^{-1/2}$$

be the normalized Laplacian of the soft adjacency matrix defined above. Then there exists a finite constant $C_{\mathcal{L}} > 0$ such that

$$\|\mathcal{L}(A) - \mathcal{L}(B)\|_2 \leq C_{\mathcal{L}} \|A - B\|_F$$

for any two admissible symmetric soft adjacency matrices A and B . Consequently,

$$\|e^{-\gamma \mathcal{L}(A)} - e^{-\gamma \mathcal{L}(B)}\|_2 \leq \gamma C_{\mathcal{L}} \|A - B\|_F.$$

Therefore, under [Theorem 2](#),

$$\mathbb{E} \|\mathbf{H}^{(t)} - \mathbf{H}^{(t-1)}\|_2 \rightarrow 0.$$

Proof. For the soft adjacency defined above, $u_{ij}^{(t)}, \tau^{(t)} \in [-1, 1]$ and $r_t \geq \epsilon_{\text{IQR}}$. Hence

$$A_{ij}^{(t)} \geq \sigma\left(-\frac{2}{\epsilon_{\text{IQR}}}\right) > 0, \quad i \neq j,$$

thus every degree is uniformly bounded below by

$$d_{\min} = (N-1) \sigma\left(-\frac{2}{\epsilon_{\text{IQR}}}\right) > 0,$$

and is bounded above by $N-1$. Let $R_A = D(A)^{-1/2}$. For two admissible matrices A and B ,

$$\|D(A) - D(B)\|_2 \leq \sqrt{N} \|A - B\|_F.$$

Since $x \mapsto x^{-1/2}$ is Lipschitz on $[d_{\min}, N-1]$,

$$\|R_A - R_B\|_2 \leq \frac{\sqrt{N}}{2d_{\min}^{3/2}} \|A - B\|_F.$$

Using $R_A A R_A - R_B B R_B = R_A (A - B) R_A + (R_A - R_B) B R_A + R_B B (R_A - R_B)$ and $\|B\|_2 \leq \|B\|_F \leq \sqrt{M}$ gives

$$\|\mathcal{L}(A) - \mathcal{L}(B)\|_2 \leq \left(\frac{1}{d_{\min}} + \frac{\sqrt{NM}}{d_{\min}^2} \right) \|A - B\|_F,$$

thus one may take

$$C_{\mathcal{L}} = \frac{1}{d_{\min}} + \frac{\sqrt{NM}}{d_{\min}^2}.$$

Finally, the Duhamel formula gives

$$e^{-\gamma \mathcal{L}(A)} - e^{-\gamma \mathcal{L}(B)} = -\gamma \int_0^1 e^{-\gamma(1-s)\mathcal{L}(A)} (\mathcal{L}(A) - \mathcal{L}(B)) e^{-\gamma s \mathcal{L}(B)} ds.$$

Both exponential factors have spectral norm at most one by [Proposition 4](#). Taking norms establishes the stated heat-kernel continuity bound. The final convergence statement follows from [Theorem 2](#). $\square$

Proposition 6 (Graph-Smoothing Effect). *Let $\mathcal{L}^{(t)} \geq 0$ and $\mathbf{H}^{(t)} = \exp(-\gamma \mathcal{L}^{(t)})$. For any $X \in \mathbb{R}^{N \times d}$ and $\tilde{X} = \mathbf{H}^{(t)} X$, the Dirichlet energy satisfies*

$$\text{Tr}(\tilde{X}^\top \mathcal{L}^{(t)} \tilde{X}) \leq \text{Tr}(X^\top \mathcal{L}^{(t)} X).$$

Proof. Let

$$\mathcal{L}^{(t)} = Q \Lambda Q^\top, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_N), \quad \lambda_k \geq 0,$$

and define $Y = Q^\top X$. Then

$$Q^\top \tilde{X} = e^{-\gamma \Lambda} Y.$$

Using cyclic invariance of the trace,

$$\text{Tr}(\tilde{X}^\top \mathcal{L}^{(t)} \tilde{X}) = \text{Tr}(Y^\top \Lambda e^{-2\gamma \Lambda} Y).$$

Because $\lambda_k e^{-2\gamma \lambda_k} \leq \lambda_k$ for every k ,

$$\Lambda e^{-2\gamma \Lambda} \leq \Lambda.$$

Therefore,

$$\begin{aligned} \text{Tr}(Y^\top \Lambda e^{-2\gamma \Lambda} Y) &\leq \text{Tr}(Y^\top \Lambda Y) \\ &= \text{Tr}(X^\top \mathcal{L}^{(t)} X). \end{aligned}$$

$\square$

Remark. ATFA has two distinct stability properties. First, [Theorem 2](#) establishes vanishing round-to-round variation of the learned soft topology, while [Proposition 5](#) transfers this regularity to the heat-diffusion operator. Second, [Propositions 4](#) and [6](#) show that, for a fixed round, heat diffusion is non-expansive and reduces graph Dirichlet energy. These results support the topology-aware aggregation mechanism without claiming more than the assumptions establish.

3.6. Computational complexity analysis

Let $M_t = |S_t|$ denote the number of clients participating in communication round t , let d denote the number of model parameters. The computational cost of ATFA consists of historical trend updating, topology construction, heat diffusion, and global aggregation. Updating the historical trend representations of all participating clients requires $\mathcal{O}(M_t d)$ operations. Constructing the pairwise cosine similarity matrix requires $\mathcal{O}(M_t^2 d)$ operations and $\mathcal{O}(M_t^2)$ additional memory. The density-calibrated threshold, adjacency matrix, and normalized graph Laplacian are subsequently computed from the similarity matrix. Computing the empirical quantiles costs $\mathcal{O}(M_t^2 \log M_t)$ due to sorting, and constructing both the adjacency matrix and the normalized Laplacian requires $\mathcal{O}(M_t^2)$. In contrast, performing the matrix exponential on the dense $M_t \times M_t$ Laplacian is substantially more expensive, with $\mathcal{O}(M_t^3)$ time complexity and $\mathcal{O}(M_t^2)$ space complexity. Applying the resulting heat kernel to the client parameter matrix $\hat{\Theta}^{(t)} \in \mathbb{R}^{M_t \times d}$ further requires $\mathcal{O}(M_t^2 d)$ operations. Therefore, the dominant server-side complexity of the exact ATFA implementation in each communication round is $\mathcal{O}(M_t^2 d + M_t^3 + M_t^2 \log M_t)$, which can be summarized as $\mathcal{O}(M_t^2 d + M_t^3)$ when the model dimension is large. The corresponding memory complexity is $\mathcal{O}(M_t d + M_t^2)$.

Table 2

Noise-injection settings. In the additive scenario, the benign update Δw is corrupted by additive noise. In the replacement scenario, the benign update is discarded and replaced entirely by noise. We denote by $r = \|\Delta w\|_2$ the norm of the benign client update. $\mathcal{N}(0, \sigma^2)$ denotes Gaussian noise with zero mean and variance σ^2 , and $\mathcal{U}(-a, a)$ denotes uniform noise over the interval $[-a, a]$.

Type	Update Rule	Noise Parameter
Gaussian Additive	$\Delta w' = \Delta w + \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$	$\sigma = s \cdot r$
Uniform Additive	$\Delta w' = \Delta w + \epsilon, \epsilon \sim \mathcal{U}(-a, a)$	$a = s \cdot r$
Gaussian Replace	$\Delta w' = \epsilon, \epsilon \sim \mathcal{N}(0, \sigma^2)$	$\sigma = s \cdot r$
Uniform Replace	$\Delta w' = \epsilon, \epsilon \sim \mathcal{U}(-a, a)$	$a = s \cdot r$

4. Experiments

4.1. Experimental setup

All experiments in this study were conducted using Python 3.9.7 and PyTorch 1.10.2.

Model. Each client trains a lightweight CNN with two convolutional layers, as is standard in non-IID FL benchmarks ([McMahan et al., 2016](#)).

Federated Settings. (1) Number of clients (N): Default value is 10. We set 50 to evaluate the scalability performance ([Table 6](#)). (2) Participation ratio (C): Default value is 1.0 (full participation). We also investigate the partial participation scenario in the scalability analysis. (3) Local epochs (E): Fixed at 5. (4) Batch size (B): Default value is 64 for all clients. (5) Optimizer: We use SGD with a learning rate of 0.01 for the benchmark datasets and 0.001 for the medical imaging benchmark datasets. (6) Communication rounds (T): Default value is 100, which is sufficient for convergence in all compared methods under our settings. All methods are evaluated using the mean best test accuracy and its standard deviation. To mitigate the potential bias caused by random initialization and improve reproducibility, we ran each experiment three times with seeds $\{1, 17, 42\}$.

Dataset. We conducted a systematic evaluation of the proposed method on two types of datasets, including standard benchmark datasets and medical imaging benchmark datasets. The standard datasets contain CIFAR-10 ([Krizhevsky, 2009](#)), CIFAR-100 ([Krizhevsky, 2009](#)), and Fashion-MNIST (FMNIST) ([Xiao et al., 2017](#)), while the medical imaging benchmark datasets include OrganAMNIST, and OrganCMNIST ([Yang et al., 2021](#)).

Dataset partition. We evaluate ATFA under both practical and pathological non-IID settings. For the practical non-IID setting, the training samples of each class are distributed across clients according to a Dirichlet distribution. We consider two experimental scenarios with Dirichlet concentration parameters set to 0.1 and 0.5, respectively. For the pathological non-IID setting ([Ye et al., 2023](#)), denoted as NIID2, the training samples are first sorted by class and divided into class-specific shards. Each non-IID client is assigned samples from two classes, resulting in highly restricted local label distributions. This protocol creates substantial distribution differences among clients and is used to evaluate the robustness of ATFA under extreme statistical heterogeneity.

An additional robustness evaluation is conducted under model update noise, as detailed in [Table 2](#). The default noise scale is set to $s = 1.0$, and 20% of the participating clients are selected.

Baseline methods. We compare ATFA with nine representative baseline methods, which can be categorized as follows:

- **Classical FL:** FedAvg ([McMahan et al., 2016](#));
- **Local model adjustment:** FedProx ([Li et al., 2020](#)), MOON ([Li et al., 2021](#)), and FedDyn ([Acar et al., 2021](#));
- **Global model adjustment:** FedNova ([Wang et al., 2020](#));
- **Personalized and topology-aware FL:** FedDC ([Gao et al., 2022](#)), FedUC ([Ma et al., 2024b](#)), FedGraph ([Chen et al., 2021a](#)), and pFedGAT ([Zhou et al., 2025b](#)).

To facilitate stable convergence of the baseline methods, their specific hyperparameters were configured according to the settings reported or recommended in the corresponding original publications. FedProx used a proximal coefficient (μ) of 0.01 and a regularization coefficient (L_2) of 10^{-5} . MOON used a contrastive coefficient (μ) of 0.01. FedDyn set the dynamic regularization coefficient to 10^{-2} . FedDC set the local correction coefficient to 10^{-2} . FedUC used a strong convexity coefficient (μ) of 0.01, and a smoothness constant (L) of 1.0. FedGraph set the graph regularization weight to 0.5. pFedGAT used a server-side attention representation dimension of 64, eight attention heads, a LeakyReLU negative slope of 0.2, a graph attention network learning rate of 0.001, and a feedback sharpening coefficient of 1.0.

4.2. Experimental results

4.2.1. Comparison with state-of-the-art baseline methods

To further evaluate the effectiveness and stability of ATFA, we systematically compare it with nine representative FL methods across five datasets and three heterogeneous data distributions. As reported in Table 3, ATFA achieves competitive overall performance and delivers notable improvements. Its most consistent advantages are observed on CIFAR-10. ATFA also performs favorably on CIFAR-100 across the three distributions. On FMNIST, ATFA obtains the best result under NIID2, exceeding FedUC and FedGraph by 0.22% and 0.20%, respectively, while ranking second under Dir(0.1). On OrganAMNIST, ATFA achieves the highest accuracy under Dir(0.5), although it slightly trails FedUC under NIID2, it remains highly competitive. Similarly, ATFA shows strong performance on OrganCMNIST, where it surpasses all baselines across all three heterogeneity settings. The convergence curves in Figs. 3 and 4 further confirm that ATFA achieves faster convergence across both standard and medical imaging datasets under various non-IID conditions. On the CIFAR-10 dataset, ATFA converges rapidly and stabilizes within 50 communication rounds. In contrast, the baseline methods exhibit slower convergence. For the more challenging CIFAR-100, ATFA maintains a clear advantage in convergence speed throughout the training process. Moreover, ATFA achieves faster convergence in most scenarios while exhibiting similarly robust performance on the medical imaging datasets.

The observed performance gains are attributed to the adaptive topology-aware aggregation mechanism. The Laplacian-based heat diffusion mitigates optimization divergence by balancing the preservation of client-specific information with collaborative knowledge fusion. These designs not only improve convergence stability but also significantly enhance the generalization capability of the global model by capturing underlying structural relationships.

4.2.2. Statistical analysis

To determine whether the performance gains of ATFA are statistically consistent across experimental settings, we performed Wilcoxon signed-rank tests against each baseline and applied Holm's correction for multiple comparisons. We further reported the mean paired accuracy differences with 95% confidence intervals and adopted a significance level of 0.05. As shown in Table 4, all Holm-adjusted p -values indicate that the performance differences between ATFA and all nine competing methods are statistically significant. ATFA demonstrates particularly clear advantages over the conventional FL baselines. It outperforms FedAvg and FedProx in all 15 settings. Compared with FedDyn, FedDC, and FedGraph, ATFA achieves 13, 15, and 15 wins, respectively. Their Holm-adjusted p -values provide strong evidence that the improvements are not attributable to random variation across experimental settings. Nevertheless, ATFA achieves 14 wins against FedUC. These results provide a statistically significant overall advantage when performance differences across all datasets and heterogeneity conditions are jointly considered. Meanwhile, the variation in the W/T/L shows that ATFA exhibits decisive gains over baselines and more modest but still statistically significant improvements over the strongest competitors.

4.2.3. Parameter sensitivity analysis

Infusion coefficient λ . To evaluate the impact of the infusion coefficient λ , we conduct a parameter sensitivity study on CIFAR-10 and FMNIST under Dir(0.5). Experimental results in Fig. 5(a) indicate that the optimal λ is closely tied to task complexity and local update stability. For the complex CIFAR-10 dataset, the performance reaches its peak at $\lambda = 0.1$ and subsequently degrades markedly with increasing λ . This suggests that excessive infusion of historical trend information may introduce excessive historical bias and reduce responsiveness to current-round updates, ultimately obscuring the critical instantaneous updates required for optimal aggregation. In contrast, FMNIST exhibits a broader favorable range ($\lambda \in [0.1, 0.3]$), as historical trend information can align more closely with current updates. However, even on FMNIST, a sharp decline occurs at $\lambda = 0.55$, further confirming that excessive reliance on historical trend information may reduce the effect of the collaboration structure. These results suggest that ATFA performs optimally with a relatively small λ value, while appropriate historical trend information remains beneficial for robust similarity measurement. To achieve optimal model performance, $\lambda \in [0.05, 0.3]$ provides a reasonable search range, with a preferred value of 0.1 specifically for complex heterogeneous tasks.

Heat diffusion γ . The sensitivity to the heat diffusion parameter γ is investigated over $\{0, 0.1, 0.25, 0.5, 1, 2, 4\}$, as shown in Fig. 5(b). On FMNIST, ATFA exhibits relatively low sensitivity to the diffusion strength. The highest accuracy is obtained at $\gamma = 2$, which is 0.1% higher than the accuracy obtained at $\gamma = 0$. Meanwhile, the limited variation among most parameter settings suggests that ATFA remains robust over a relatively broad range of diffusion strengths. A more evident sensitivity is observed on CIFAR-10. The average best accuracy ranges from 71.25% to 72.22%, resulting in a maximum variation of 0.97%. Similar to FMNIST, the best performance is achieved at $\gamma = 2$. The accuracies obtained at $\gamma = 0.5$ and $\gamma = 1$ indicate that several moderate diffusion strengths provide competitive performance. Increasing the diffusion strength does not continuously improve the performance of the global model. A relatively small γ may provide insufficient knowledge fusion over the learned graph, whereas an excessively large value may over-diffuse client representations and reduce the distinction between heterogeneous local updates.

4.2.4. Inter-client similarity measurement analysis

In this subsection, we evaluate two distinct similarity measures within the proposed ATFA method. The first strategy is the accumulated gradients method (AGM). The second strategy is the proposed historical trend information (HTI). As illustrated in Fig. 6, the overall experimental results indicate that these two strategies experience a rapid increase in accuracy during the early training phase, followed by a gradual convergence toward stability. HTI presents superior performance during the mid-to-late stages, while the convergence curve of AGM exhibits significant fluctuations after 60 rounds. These fluctuations suggest that AGM is more sensitive to round-specific gradient variation and lead to less stable convergence of the global model. In contrast, HTI can effectively mitigate the impact of different local optimization directions based on the construction of graph topology. This superiority can be attributed to the incorporation of historical update trends, which directly characterize the cross-round evolution of inter-client relationships in the model manifold, rather than relying solely on accumulated gradients to correct local updates. In the presence of client-specific submanifold structures, HTI effectively synchronizes client updates that are located on the same or adjacent submanifolds, enhancing the stability of the aggregation process.

4.2.5. Ablation study

To evaluate the contribution of the adaptive graph topology and the Laplacian-based heat diffusion, we compare the complete ATFA with two variants. The first variant, ATFA without graph topology (w/o

Table 3
Accuracy comparison of the state-of-the-art baselines on the different data distributions.

		FedAvg	FedProx	FedDyn	MOON	FedNova	FedDC	FedUC	FedGraph	pFedGAT	ATFA
Fashion-MNIST	Dir(0.1)	83.35 $\pm 8.89E-03$	84.37 $\pm 1.10E-02$	86.29 $\pm 6.61E-03$	88.07 $\pm 2.07E-02$	85.44 $\pm 1.58E-02$	85.77 $\pm 2.19E-03$	85.77 $\pm 4.37E-03$	85.72 $\pm 5.90E-03$	82.84 $\pm 1.73E-02$	86.30 $\pm 5.51E-03$
	Dir(0.5)	87.84 $\pm 8.63E-03$	89.08 $\pm 5.85E-03$	89.41 $\pm 1.69E-03$	89.28 $\pm 3.30E-03$	89.23 $\pm 3.71E-03$	89.27 $\pm 1.50E-03$	89.47 $\pm 2.81E-03$	89.32 $\pm 3.00E-04$	89.12 $\pm 3.70E-03$	89.75 $\pm 2.20E-03$
	NIID2	86.37 $\pm 3.51E-03$	86.69 $\pm 6.41E-03$	87.65 $\pm 2.32E-03$	86.93 $\pm 3.30E-03$	87.27 $\pm 7.63E-03$	87.65 $\pm 1.72E-03$	87.72 $\pm 3.56E-03$	87.74 $\pm 9.00E-03$	87.02 $\pm 1.05E-02$	87.94 $\pm 8.80E-04$
CIFAR-10	Dir(0.1)	58.65 $\pm 1.72E-02$	62.32 $\pm 2.20E-02$	66.67 $\pm 7.95E-03$	53.99 $\pm 6.81E-03$	64.93 $\pm 1.36E-02$	63.16 $\pm 1.82E-02$	62.43 $\pm 1.22E-02$	61.35 $\pm 5.13E-02$	60.76 $\pm 3.89E-02$	69.07 $\pm 3.70E-03$
	Dir(0.5)	63.70 $\pm 3.89E-02$	66.65 $\pm 1.34E-02$	70.25 $\pm 1.54E-03$	66.14 $\pm 8.47E-03$	67.12 $\pm 6.38E-03$	70.45 $\pm 3.81E-03$	67.08 $\pm 1.28E-02$	67.44 $\pm 4.10E-03$	66.21 $\pm 3.36E-02$	71.11 $\pm 4.80E-03$
	NIID2	63.66 $\pm 1.17E-02$	64.63 $\pm 2.77E-03$	66.84 $\pm 4.05E-03$	64.78 $\pm 9.71E-03$	64.12 $\pm 2.88E-02$	67.20 $\pm 6.64E-03$	65.12 $\pm 4.95E-04$	65.96 $\pm 2.90E-03$	64.38 $\pm 1.27E-02$	69.92 $\pm 5.80E-03$
CIFAR-100	Dir(0.1)	24.02 $\pm 2.52E-02$	27.59 $\pm 8.60E-03$	31.04 $\pm 3.92E-03$	27.85 $\pm 6.09E-03$	28.10 $\pm 1.60E-02$	30.50 $\pm 1.16E-02$	28.71 $\pm 8.41E-03$	28.08 $\pm 3.20E-03$	25.37 $\pm 2.74E-02$	33.36 $\pm 7.00E-04$
	Dir(0.5)	27.42 $\pm 2.37E-02$	31.60 $\pm 1.02E-02$	34.52 $\pm 2.29E-03$	31.55 $\pm 1.11E-02$	31.51 $\pm 1.61E-02$	34.30 $\pm 2.37E-03$	32.35 $\pm 2.57E-03$	31.87 $\pm 3.30E-03$	31.33 $\pm 1.46E-02$	34.90 $\pm 3.90E-03$
	NIID2	24.49 $\pm 7.87E-03$	28.82 $\pm 7.75E-03$	31.98 $\pm 3.57E-03$	29.07 $\pm 7.62E-03$	28.89 $\pm 2.59E-02$	32.00 $\pm 1.20E-03$	30.48 $\pm 7.17E-03$	29.95 $\pm 6.30E-03$	27.68 $\pm 1.50E-02$	32.28 $\pm 1.60E-03$
OrganAMNIST	Dir(0.1)	41.71 $\pm 9.60E-03$	77.76 $\pm 2.00E-04$	79.58 $\pm 2.20E-02$	77.72 $\pm 2.00E-04$	78.91 $\pm 4.00E-04$	77.95 $\pm 2.10E-03$	77.79 $\pm 8.00E-04$	76.40 $\pm 2.02E-02$	79.30 $\pm 3.77E-02$	79.01 $\pm 1.04E-02$
	Dir(0.5)	51.65 $\pm 1.25E-02$	77.82 $\pm 9.00E-04$	80.87 $\pm 6.70E-03$	76.85 $\pm 1.49E-02$	78.13 $\pm 3.00E-04$	80.34 $\pm 3.00E-04$	77.73 $\pm 7.00E-04$	76.89 $\pm 1.58E-02$	79.36 $\pm 3.73E-02$	82.26 $\pm 5.10E-03$
	NIID2	45.41 $\pm 5.48E-02$	77.79 $\pm 1.00E-03$	80.55 $\pm 1.44E-02$	76.65 $\pm 1.75E-02$	77.75 $\pm 1.00E-04$	80.07 $\pm 1.27E-02$	80.72 $\pm 2.64E-02$	76.87 $\pm 1.54E-02$	79.37 $\pm 3.72E-02$	80.35 $\pm 8.80E-03$
OrganCMNIST	Dir(0.1)	65.76 $\pm 2.48E-02$	73.11 $\pm 5.86E-02$	81.40 $\pm 4.20E-03$	72.32 $\pm 6.95E-02$	79.61 $\pm 9.84E-02$	82.97 $\pm 2.80E-03$	83.77 $\pm 1.80E-03$	84.18 $\pm 6.60E-03$	83.45 $\pm 5.00E-03$	84.26 $\pm 1.00E-04$
	Dir(0.5)	65.95 $\pm 2.20E-02$	70.21 $\pm 8.28E-02$	82.49 $\pm 3.70E-03$	72.24 $\pm 6.88E-02$	78.04 $\pm 1.18E-01$	83.06 $\pm 7.80E-03$	83.93 $\pm 2.80E-03$	84.31 $\pm 7.30E-03$	83.53 $\pm 5.50E-03$	84.39 $\pm 2.30E-03$
	NIID2	65.98 $\pm 2.24E-02$	72.13 $\pm 6.78E-02$	83.38 $\pm 2.30E-03$	72.27 $\pm 6.90E-02$	65.80 $\pm 1.97E-02$	83.52 $\pm 9.50E-03$	84.41 $\pm 5.20E-03$	84.30 $\pm 6.50E-03$	83.35 $\pm 5.10E-03$	84.43 $\pm 1.60E-03$

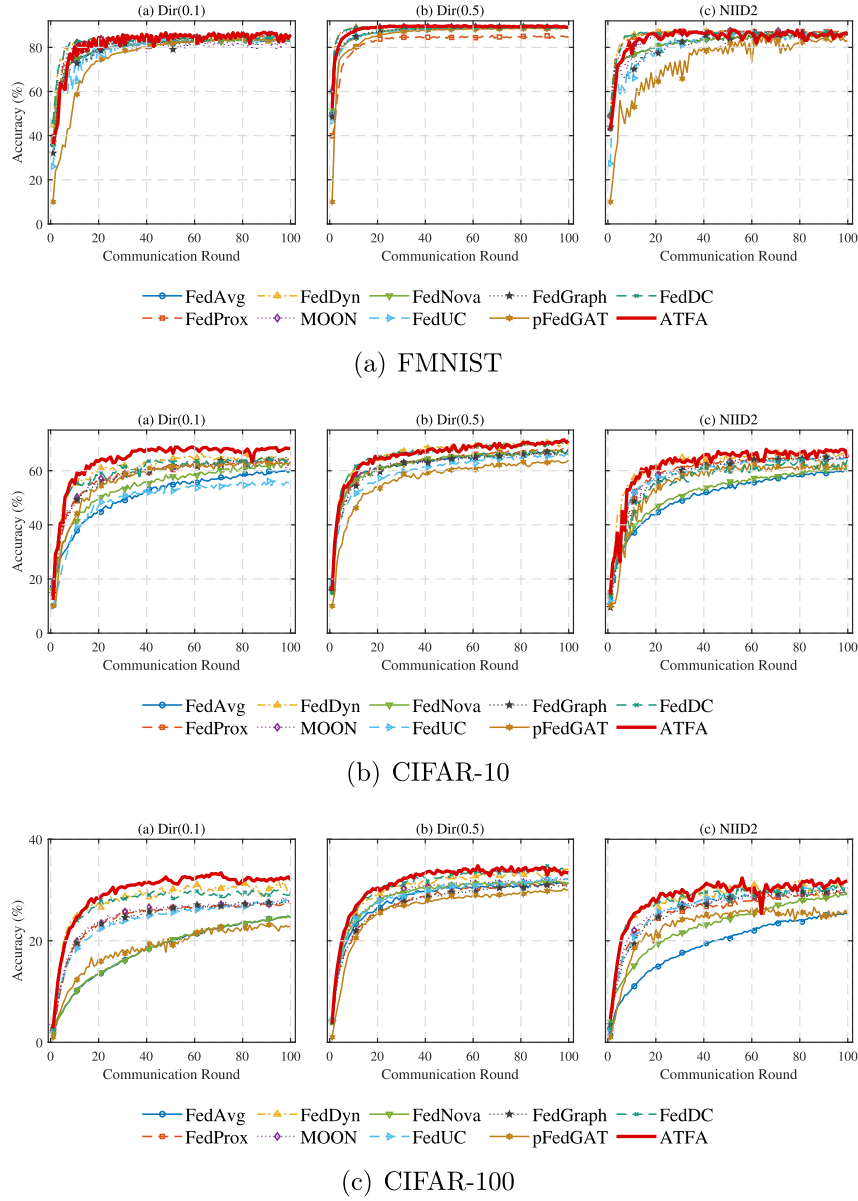


Fig. 3. Convergence curves under different data distributions on the benchmark datasets.

Table 4

Statistical test of ATFA and baseline methods across all dataset heterogeneity settings.

Baseline	W/T/L	Mean Δpp	95% CI	Raw p	Holm p	Sig.
FedAvg	15/0/0	14.255	[8.858,20.295]	6.1035E-05	5.4932E-04	Yes
FedProx	15/0/0	5.281	[3.363,7.424]	6.1035E-05	5.4932E-04	Yes
FedNova	15/0/0	4.33	[2.551,6.752]	6.1035E-05	5.4932E-04	Yes
FedDC	15/0/0	1.439	[0.819,2.243]	6.1035E-05	5.4932E-04	Yes
FedGraph	15/0/0	2.627	[1.528,3.798]	6.1035E-05	5.4932E-04	Yes
pFedGAT	14/0/1	3.115	[1.861,4.482]	1.2207E-04	5.4932E-04	Yes
MOON	14/0/1	5.606	[3.281,8.120]	4.2725E-04	1.2817E-03	Yes
FedUC	14/0/1	2.155	[1.118,3.279]	4.2725E-04	1.2817E-03	Yes
FedDyn	13/0/2	1.125	[0.578,1.695]	1.5259E-03	1.5259E-03	Yes

GT), disables the topology-aware client-relationship module. The second variant, ATFA without heat diffusion (w/o HD), retains the adaptive graph topology but removes the heat diffusion module. It can be seen in Fig. 7 that the complete ATFA achieves the best performance across all datasets and heterogeneity settings, which confirms the effectiveness of the adaptive graph topology and the heat diffusion. On FMNIST, the advantage of the full ATFA is relatively moderate. Under Dir(0.1), re-

moving the heat diffusion module causes a performance drop of 0.7%, while removing graph topology leads to a larger drop of 3.5%. A similar trend is observed under Dir(0.5), w/o HD and w/o GT lead to decreases of 0.4% and 1.9%, respectively. A more pronounced difference is observed on CIFAR-10. Under Dir(0.5), the full ATFA clearly outperforms w/o HD and w/o GT. Under Dir(0.1), the effect becomes even more pronounced. Notably, w/o HD performs even slightly worse than

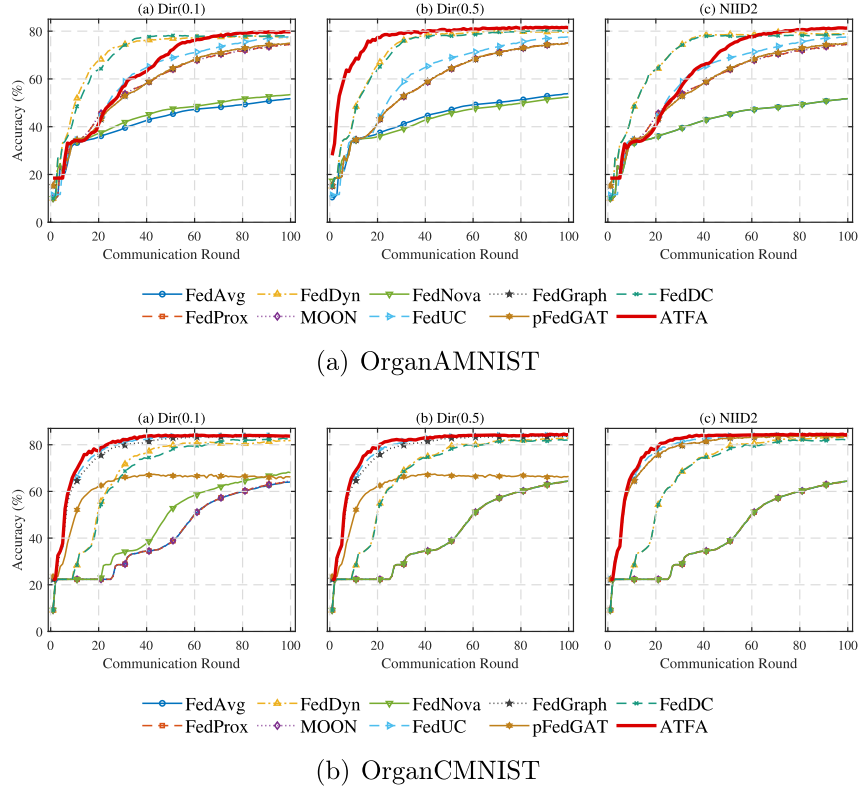


Fig. 4. Convergence curves under different data distributions on the medical imaging datasets.

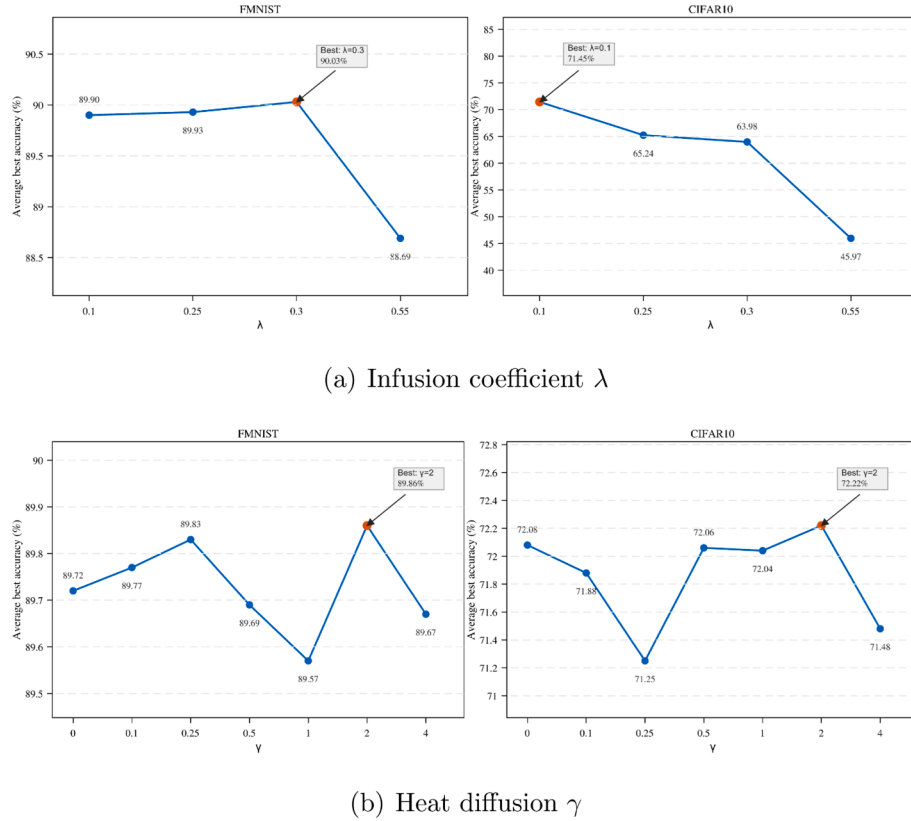


Fig. 5. Sensitivity analysis of the parameter λ and γ on FMNIST and CIFAR-10 datasets under Dir(0.5).

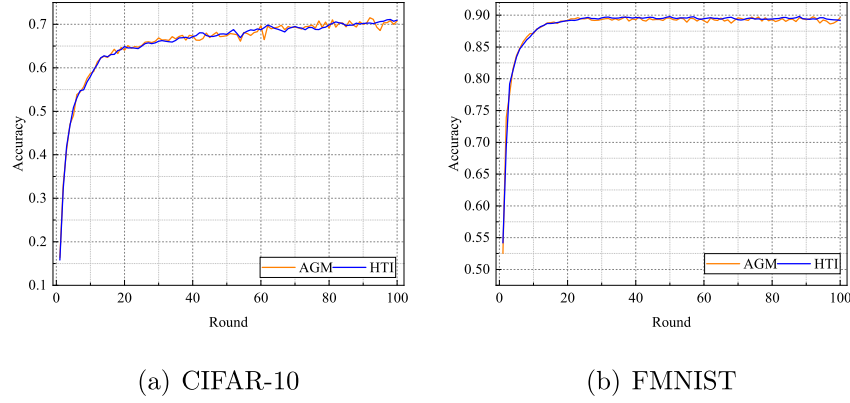


Fig. 6. Effects of accumulated gradients method (AGM) and historical trend information (HTI) under Dir(0.5).

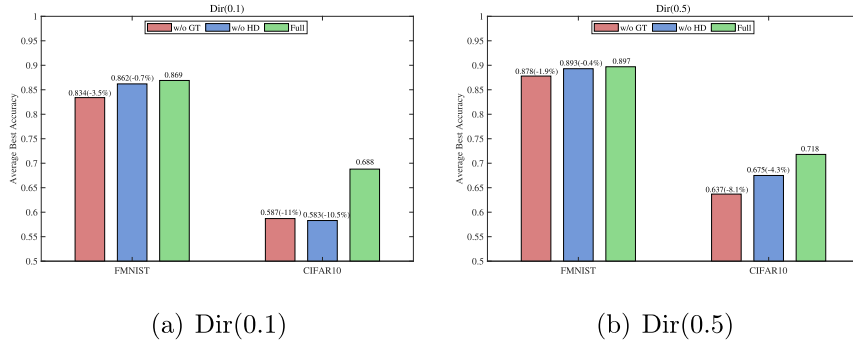


Fig. 7. Ablation results of ATFA on FMNIST and CIFAR-10.

w/o GT, suggesting that adaptive topology construction alone is insufficient to fully exploit inter-client relationships without subsequent heat diffusion-based knowledge fusion. The adaptive graph topology explicitly characterizes structural dependencies among client updates, thereby reducing indiscriminate information fusion during aggregation. On this basis, the heat diffusion propagates model information according to the learned graph structure and promotes knowledge fusion among strongly related clients.

4.2.6. Analysis of topology-aware knowledge fusion

To investigate the stability of the graph topology, we conduct an experiment involving graph perturbation interventions. Specifically, at designated communication rounds ($T=10, 20, 30$, and 40), we systematically remove edges from the client graph and monitor the subsequent changes in test accuracy and short-term recovery behavior. The results presented in Fig. 8 provide empirical evidence that the constructed topology promotes effective collaboration among clients. Despite the edge-removal perturbations, the test accuracy of ATFA exhibits a consistent upward trend on both FMNIST and CIFAR-10, suggesting robustness to the specified edge-removal interventions. The recovery behavior substantiates the functional role of graph topology and is associated with considerable gains in recovery, such as a 2.42% improvement at T_{10} on CIFAR-10. In contrast, the diminishing returns observed in later rounds (e.g., at T_{40}) indicate that convergence has been achieved. These results confirm that the Laplacian-based heat diffusion significantly facilitates inter-client knowledge fusion through the adaptive graph topology. Moreover, as illustrated in Fig. 9, strongly connected client pairs consistently exhibit higher update alignment than weakly connected pairs in both scenarios. The observed neighbor alignment gain, which is consistent with the learned graph encoding structurally aligned client relationships, indicates a sustained enhancement in spatial consistency among neighboring clients. These results further confirm that ATFA provides a discriminative characterization of client relationships and continuously

strengthens the collaborative direction among clients with similar structures.

To further examine the topology-aware knowledge fusion, we introduce additional graph-control and edge-intervention experiments. As shown in Table 5, the learned graph consistently achieves the highest accuracy on both datasets. On FMNIST, although the discrepancies are relatively small, the learned graph still consistently outperforms all graph settings. The advantage of the learned topology is substantially more pronounced on CIFAR-10. The correspondence between learned edge weights and specific client pairs provides relevant structural information for topology-aware knowledge fusion. The fact that the random graph still slightly outperforms the no-graph settings suggests that a general graph may offer a limited regularization effect. However, the substantially larger improvement obtained by the learned graph highlights the importance of topology quality. On CIFAR-10, the larger performance degradation caused by removing high-weight edges indicates that these edges encode more influential collaborative relationships than low-weight edges. FMNIST exhibits a different pattern, with no clear performance ordering observed between high- and low-weight edge removal. The superiority of the learned graph over both random and no-graph shows that the improvement is associated with informative client-specific topology rather than the mere existence of graph-based propagation. Moreover, the edge-intervention results on CIFAR-10 indicate that the learned weights contain information about differentiated edge importance.

4.2.7. Analysis of adaptive graph topology evolution

To further investigate the evolution of learned client relationships over the course of training, we examine the weighted adjacency matrices at representative communication rounds. As shown in Fig. 10, these matrices are visualized for rounds 1, 20, 50, and 100. The adjacency matrices show that the learned topology remains effectively sparse under the stated edge-selection rule throughout training while adaptively

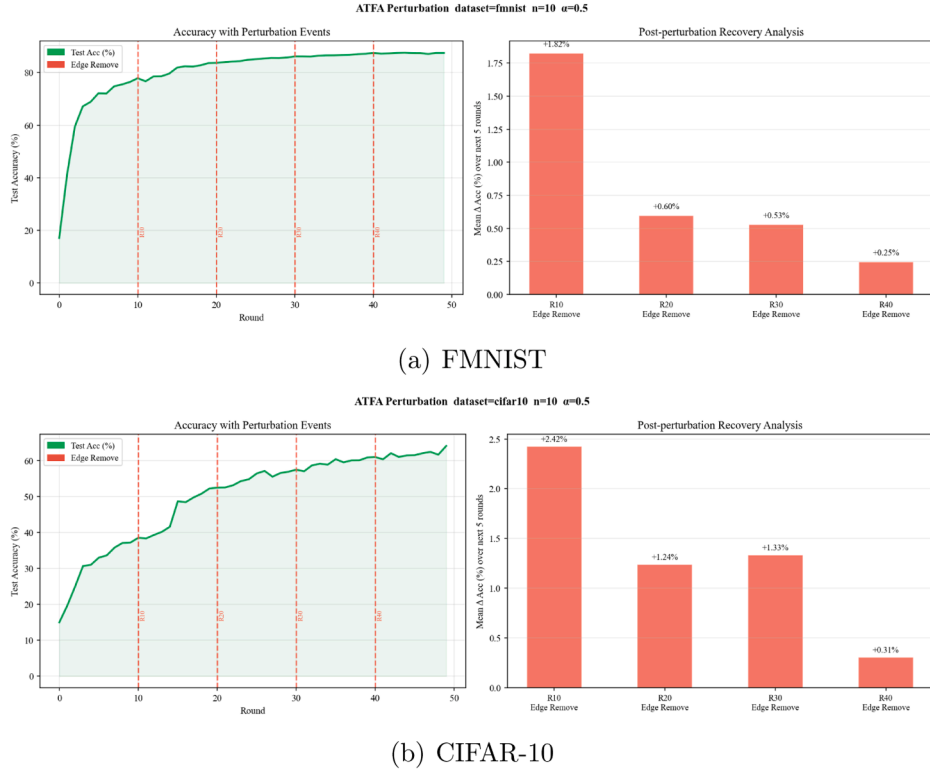


Fig. 8. Graph perturbation intervention analysis of ATFA under Dir(0.5).

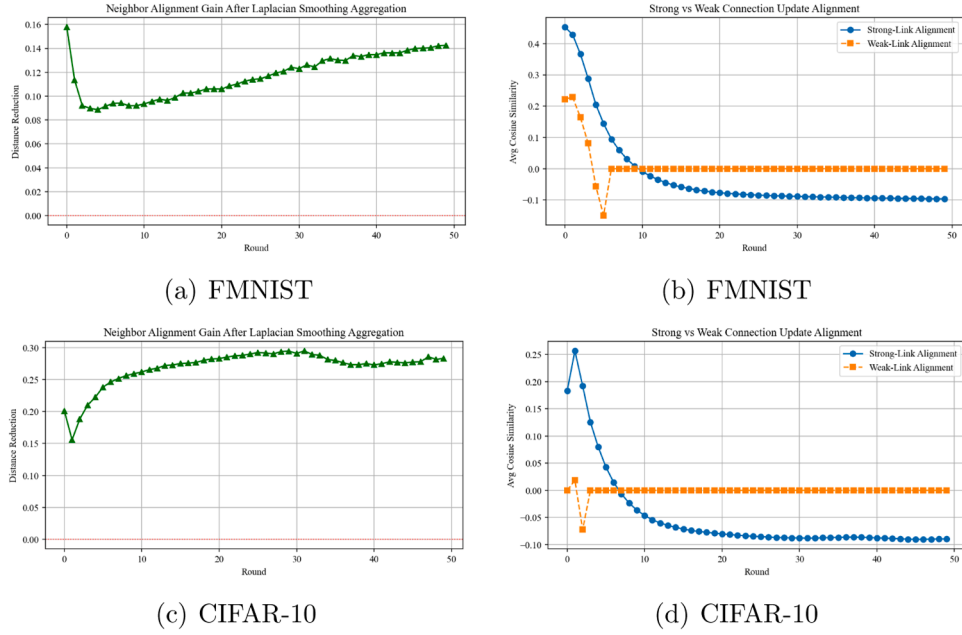


Fig. 9. Evolution of topology-aware update alignment on FMNIST and CIFAR-10. We report (i) the update alignment between strongly and weakly connected client pairs over communication rounds, and (ii) the neighbor alignment gain.

Table 5

Effects of graph settings and edge-weight interventions on FMNIST and CIFAR-10 under Dir(0.5).

Graph setting	FMNIST		CIFAR-10	
	Best average accuracy (%)	Degradation vs. learned graph (pp)	Best average accuracy (%)	Degradation vs. learned graph (pp)
Learned graph (Ours)	89.9	-	72.21	-
Random graph	89.6	0.30%	67.12	5.09%
No graph	89.32	0.58%	65.92	6.29%
Remove low-weight edges(20%)	89.59	0.31%	68.08	4.13%
Remove high-weight edges(20%)	89.75	0.15%	67.57	4.64%

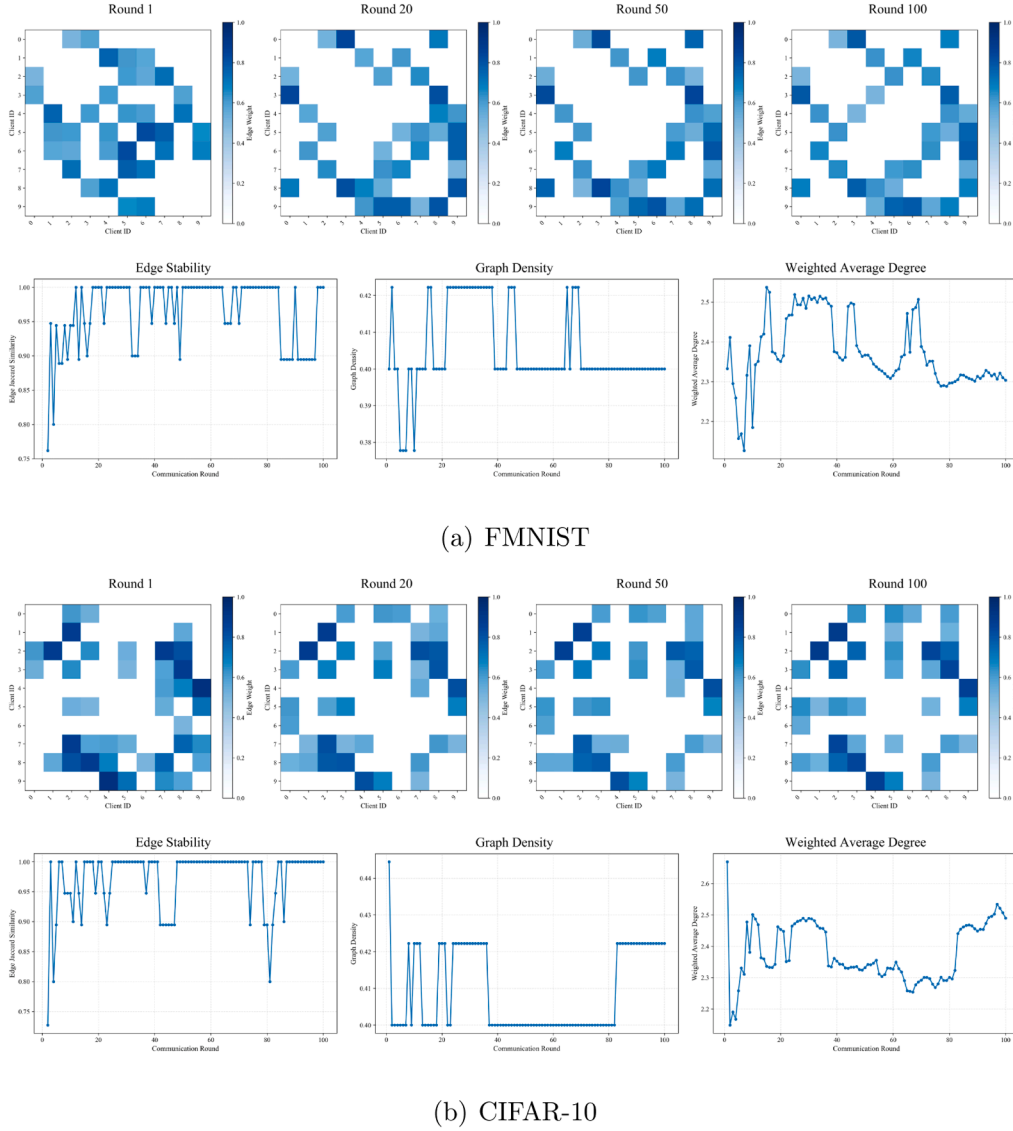


Fig. 10. Evolution of the learned adaptive graph topology on (a) FMNIST and (b) CIFAR-10 under Dir(0.5).

updating both the set of active edges and their corresponding weights across communication rounds. The graph degenerates into neither an isolated structure nor a fully connected one. This behavior indicates that the proposed density-calibrated construction mechanism preserves informative inter-client dependencies without indiscriminately connecting all clients. On FMNIST, the topology undergoes relatively pronounced adjustments during the initial communication rounds. This suggests that the client relationships constructed from the limited historical information available at the beginning of training are progressively refined as additional optimization trajectories are accumulated. The temporary increase in the weighted average degree may reflect stronger connectivity among clients with increasingly aligned historical optimization trajectories, thereby supporting more intensive knowledge fusion during this stage of training. Notably, reductions in edge stability without corresponding changes in graph density indicate that edges are being substituted under a nearly constant connectivity budget rather than that the graph structure is collapsing. A similar but more dynamic evolution can be observed on CIFAR-10. Following the initial adaptation stage, edge stability remains close to 1.0 for most communication rounds, demonstrating substantial short-term continuity in the learned relationships. The concurrent change in edge stability and density indicates a structural transition in which a limited number of client relationships are

reorganized. The observed late-phase improvement in performance corresponds to the increasing graph density, indicating that the server recognizes strengthened collaborative ties as client representations become more informative. Compared with FMNIST, CIFAR-10 exhibits more evident late-stage topology reconfiguration, which may be associated with its more complex optimization geometry and inter-client representation relationships.

4.2.8. Robustness against model update noise

We perform experiments on two benchmark datasets under various noise perturbations to evaluate the robustness of ATFA. Fig. 11 summarizes the comparative results, showing that FedAvg without noise remains inferior to ATFA under noisy conditions. Notably, the performance gain is more pronounced on CIFAR-10 (+5.02%) compared to FMNIST (+0.50%). The preservation of historical trend information mitigates the influence of high-frequency random noise. The design of robust dispersion statistics in the construction of the soft adjacency can improve the resilience of the graph topology. Furthermore, the knowledge fusion process can mitigate local divergence and preserve essential global features by guiding adjacent clients with similar representations to converge toward a unified optimization direction.

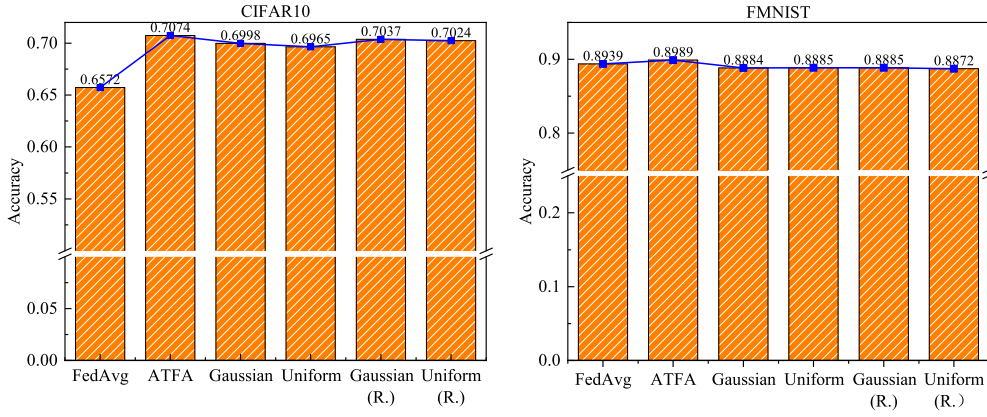


Fig. 11. Accuracy comparison under different noise-injection strategies on CIFAR-10 and FMNIST.

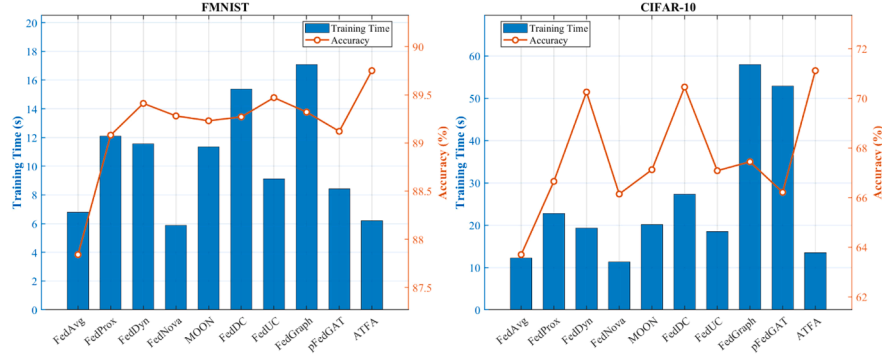


Fig. 12. Comparison of computational time on FMNIST and CIFAR-10 under Dir(0.5).

Table 6

Evaluation of scalability across different settings under Dir(0.5).

Sample Fraction		FedAvg	FedProx	FedDyn	FedNova	MOON	FedDC	FedUC	FedGraph	pFedGAT	ATFA
0.8	Mean	60.29%	62.78%	63.76%	62.81%	62.05%	65.13%	61.34%	65.39%	64.35%	68.83%
	Std.	1.40E-02	5.18E-03	2.25E-03	1.08E-02	2.16E-02	4.81E-03	1.17E-02	3.03E-02	3.24E-02	1.70E-02
0.5	Mean	58.97%	60.88%	62.97%	61.88%	60.91%	65.13%	60.22%	65.03%	61.53%	66.24%
	Std.	1.75E-02	3.54E-04	8.91E-03	1.61E-02	3.39E-03	4.81E-03	3.04E-03	2.98E-02	3.69E-02	2.17E-02

4.2.9. Scalability analysis

To evaluate the robustness of ATFA under partial client participation, we conduct experiments on CIFAR-10 with client sampling fractions of 0.8 and 0.5. As shown in Table 6, ATFA achieves the highest test accuracy under both settings. With a sampling fraction of 0.8, ATFA outperforms the strongest baseline by 3.44%. When the sampling fraction is reduced to 0.5, ATFA still outperforms the strongest competing method. Although the reduced participation rate causes a decrease in ATFA accuracy, it remains superior to all baselines, demonstrating its effectiveness when only a subset of clients is available in each communication round. The corresponding standard deviations increase moderately, indicating slightly greater variability under lower participation while preserving the overall performance advantage. These results corroborate the design rationale of ATFA. Specifically, historical trends aid in reliably constructing inter-client relationships as the client subset changes, while the heat diffusion mechanism propagates information over the client graph, thereby preventing excessive dependence on locally selected clients.

4.3. Computational time analysis

Computational efficiency is evaluated under Dir(0.5), ATFA exhibits a lower observed per-round computational time, as shown in Fig. 12. Although ATFA introduces several additional modules, these operations incur only limited computational overhead. The additional computa-

tions are lightweight and performed on low-dimensional client graphs, ensuring a minimal increase in overall per-round computational time. Consequently, ATFA achieves a balance between computational efficiency and optimization performance without a significant increase in computational overhead.

Table 7 further reports the computational cost of ATFA under different numbers of participating clients on FMNIST and CIFAR-10. Across both datasets, the total diffusion time increases with the number of participating clients (N), reflecting the growing cost of constructing the client graph and applying heat diffusion to the client models. On FMNIST, the diffusion overhead remains below 5% for $N \leq 200$, but reaches 11.590% under full participation. On CIFAR-10, the average diffusion overhead remains below 5% for all evaluated client scales, including full client participation. Importantly, the exact matrix exponential is not the dominant computational bottleneck. Even in the largest setting, matrix exponentiation accounts for only approximately 0.081% of the round time on FMNIST and 0.027% on CIFAR-10. The comparatively large standard deviations observed for several small-scale CIFAR-10 settings are caused by occasional timing outliers in sub-millisecond operations, and their P95 values remain low and follow the expected increasing trend with participating clients. The growth of the complete diffusion is mainly associated with pairwise client-similarity computation and the application of the heat kernel to high-dimensional model parameters.

Table 7

Computational scalability and diffusion overhead of ATFA with varying numbers of participating clients on FMNIST and CIFAR-10 under Dir(0.5).

FMNIST							
N	Sampling Rate	Matrix size	Expm time(ms)	Diffusion time(ms)	P95 diffusion(ms)	Round time(s)	Diffusion overhead
10	0.02	10×s10	0.163±0.025	3.499±0.455	4.415	1.212±0.302	0.294±0.041
20	0.04	20×s20	0.233±0.118	9.760±1.262	11.621	1.868±0.345	0.530±0.080
50	0.1	50×s50	0.294±0.065	40.055±5.306	46.909	3.661±0.352	1.098±0.138
100	0.2	100×s100	0.619±0.151	153.460±13.849	177.872	6.644±0.350	2.314±0.221
200	0.4	200×s200	2.155±0.235	640.523±24.686	679.65	12.905±0.424	4.967±0.212
500	1	500×s500	27.128±2.715	3864.075±222.900	4276.508	33.330±1.078	11.590±0.469
CIFAR-10							
N	Sampling Rate	Matrix size	Expm time(ms)	Diffusion time(ms)	P95 diffusion(ms)	Round time(s)	Diffusion overhead
10	0.02	10×s10	0.200±1.411	7.736±5.181	8.449	2.957±0.251	0.262±0.160
20	0.04	20×s20	0.136±0.502	21.645±3.970	26.363	5.094±0.398	0.426±0.077
50	0.1	50×s50	0.308±0.464	116.981±10.978	135.788	11.196±0.591	1.046±0.093
100	0.2	100×s100	0.762±0.143	455.694±33.519	515.015	21.199±0.871	2.150±0.140
200	0.4	200×s200	2.766±0.949	1782.043±119.586	2009.468	42.134±1.432	4.228±0.220
500	1	500×s500	29.315±4.246	5190.671±331.208	5742.401	109.292±3.801	4.748±0.227

4.4. Limitations

Despite its strong performance, ATFA faces two limitations. The exact matrix exponential introduces cubic server-side complexity, which may hinder scalability to thousands of clients. Additionally, while the medical imaging benchmarks employed in this study (e.g., OrganAMNIST, OrganCMNIST) provide a standardized and reproducible framework for evaluating algorithmic performance, they may not fully capture the full complexity of real-world cross-institutional federated learning deployments. Consequently, the current experimental results should be interpreted primarily as evidence of the optimization and predictive capabilities of ATFA under controlled conditions, rather than as a direct demonstration of its clinical readiness.

5. Conclusion and future work

In this work, we propose Adaptive Topology-aware Federated Aggregation (ATFA), a novel method designed to mitigate model divergence in heterogeneous federated environments. ATFA efficiently transforms the aggregation paradigm from simple Euclidean averaging to topology-aware aggregation by capturing the intrinsic manifold structure of local model updates. The historical trend information can mitigate divergence in stochastic optimization and model evolving inter-client relationships. Additionally, we design an adaptive density-calibrated threshold that autonomously constructs a sparse and informative topology. To enhance inter-client collaboration with structural alignment, we implement topology-aware aggregation by incorporating Laplacian-based heat diffusion, which effectively suppresses interference from divergent local-model updates. Extensive evaluations on both benchmark and medical imaging datasets demonstrate that ATFA outperforms the baseline methods, supporting its effectiveness in improving the performance of the global model across the evaluated settings.

Future work will extend ATFA along several directions. To bridge the gap between benchmark evaluations and real-world clinical deployment, we plan to evaluate ATFA on institution-defined client partitions with clinically relevant criteria, and to integrate differential privacy and communication-efficient transmission protocols for practical medical federated learning settings. On the algorithmic side, we will further optimize the computational efficiency of graph evolution for ultra-large-scale federated networks through polynomial approximations (e.g., Chebyshev expansion), and explore reinforcement learning-based strategies to enhance fine-grained adaptability of the aggregation process.

CRedit authorship contribution statement

Xianqiu Meng: Writing – original draft, Methodology, Funding acquisition, Conceptualization; **Gaochao Xu:** Project administration, Methodology; **Xu Xu:** Software, Data curation; **Long Li:** Visualization, Validation, Software; **Jiaqi Ge:** Validation, Software; **Jianhua Jiang:** Writing – review & editing, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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